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Major Project Report

On

Design and Static Analysis of a Sandwich Honeycomb Structure

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Design and Static Analysis of a Sandwich Honeycomb Structure

A Report submitted in partial fulfillment of the requirements for the award of Degree of

B.Tech

*In the **Aerospace Engineering** of*
Dayananda Sagar University

by

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CERTIFICATE

This is to certify that the research work entitled **"Design and Static Analysis of a Sandwich Honeycomb Structure"** carried out by **Rohan Balram Patel (ENG21AS0037), Rithin A (ENG21AS0036), Adam Blesson (ENG22AS1001)** in partial fulfilment of the requirements for the award of degree of **B.Tech** in the **Aerospace Engineering** of **Dayananda Sagar University**, Bengaluru during the year 2024 - 2025. This work has been approved as it satisfies the academic requirements in the work prescribed for the said degree. We also state that this report doesn't contain any work which has been previously submitted for the award of any degree anywhere and is therefore a prima facie case for consideration of an award.

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DECLARATION

We, Mr. **Rohan Balram Patel (ENG21AS0037), Mr. Rithin A (ENG21AS0036), Mr. Adam Blesson (ENG22AS1001)** hereby declare that, this dissertation work entitled as **"Design and Static Analysis of a Sandwich Honeycomb Structure"** been carried out by us under the guidance of **Prof. Kartik S Tandel** Department of Aerospace Engineering, in partial fulfillment of the requirement of the degree **Bachelor of Technology in Aerospace Engineering.**

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ABSTRACT

Honeycomb core sandwich panels are widely recognized for their exceptional strength-to-weight ratio, making them ideal for various structural applications. The objective of this research is to examine the behaviour of honeycomb sandwich panel using combination of theoretical modeling and experimental validation. This project focuses the design and static structural analysis of a sandwich honeycomb structure.

The entire workflow, from 3D modeling to simulation, is executed within the ANSYS environment. The geometric modeling of the sandwich panel is carried out using ANSYS Space Claim, which allows for precise control over the dimensions and features of the honeycomb cells and face sheets. The model is then imported into ANSYS Mechanical, where static structural analysis is performed using the Finite Element Method (FEM). Material selection includes Polylactic Acid (PLA) for the honeycomb core and Carbon Fiber-Reinforced PLA for the face sheets.

In this analysis, we explore three different configurations by adjusting the thickness of the face sheets to 0.85 mm, 1.2 mm, and 1.6 mm, all while keeping the honeycomb cell geometry consistent. Each model is put through a uniformly distributed static load, and we take note of important factors like total deformation, Von Mises stress, and the safety factor.

This study emphasizes just how important face sheet thickness is when it comes to the structural efficiency of honeycomb sandwich panels. Thicker face sheets really boost stiffness, minimize deformation, and enhance load-bearing capacity under static conditions. The hexagonal core design plays a key role in distributing weight effectively while maintaining a high strength-to- weight ratio. The results show that even minor design tweaks can lead to significant performance gains. These insights highlight why sandwich structures are such a great fit for demanding weight-sensitive applications like aerospace.

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SYMBOLS/ NOMENCLATURE

P – Load applied (N)

L – Span length (mm)

b – Width of the beam (mm)

t – Face sheet thickness (mm)

d – Total thickness of the sandwich panel (mm)

c – Core thickness (mm)

E – Flexural modulus of face sheet (Pa)

G – Shear modulus of core (Pa)

τ – Core shear stress (Pa or MPa)

σ – Facing bending stress (Pa or MPa)

Δ – Midspan deflection (mm)

D – Bending stiffness ($N \cdot mm^2$)

U – Shear rigidity ($N \cdot mm$)

ρ – Relative density of honeycomb core

θ – Internal angle of hexagonal cell (typically 30°)

CHAPTER 1

1. Introduction

1.1 Problem Statement

In the pursuit of advanced engineering solutions, particularly in the fields of aerospace, automotive, marine, and civil structures, material efficiency is of utmost importance. Weight reduction without compromising structural integrity is a critical requirement for modern designs, especially in aerospace engineering, where even slight improvements in structural efficiency can lead to significant cost savings and performance enhancements [1], [2]. Traditional monolithic materials often exhibit high stiffness and strength but at the cost of increased weight. To overcome this limitation, composite structures and innovative geometrical configurations have emerged as viable alternatives [3], [4].

Sandwich structures with honeycomb cores are a notable example of such innovations. These structures consist of two thin, stiff face sheets bonded to a lightweight but thick core, typically arranged in a honeycomb pattern [1], [5]. This configuration leverages the mechanical advantages of both the face sheets and the core, resulting in a structure with high bending stiffness, strength, and excellent energy absorption characteristics while keeping the overall weight minimal [6], [7]. Honeycomb sandwich panels are extensively used in aerospace components such as aircraft fuselage panels, flooring systems, and control surfaces due to their favorable strength-to-weight ratio [2], [8]

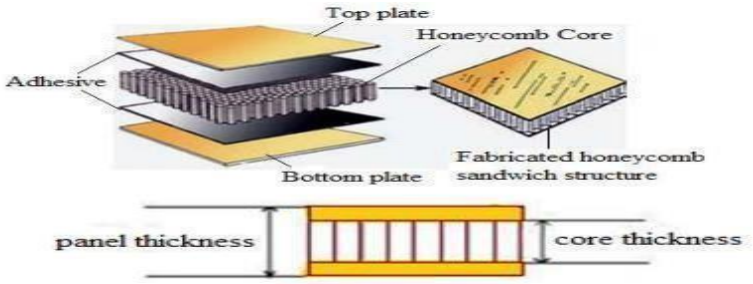


Figure 1. Exploded view of a honeycomb sandwich panel illustrating the assembly of face sheets and the internal honeycomb core. This configuration enhances flexural stiffness and energy absorption capabilities.
Source: YILDIZ Technical University

However, the challenge lies in optimizing the design of such structures to meet specific loading and functional requirements. Variations in cell size, core thickness, wall thickness, and material properties can significantly influence the mechanical performance of the panel [9]. Moreover, ensuring uniform load distribution, minimal deformation, and maximum safety under static conditions requires rigorous design and analysis [10]. This project aims to address these challenges by designing a sandwich honeycomb structure and evaluating its static performance through numerical simulations.

In particular, the use of additive manufacturing (3D printing) to fabricate honeycomb sandwich structures presents a modern and customizable approach to material design [5], [11]. By adjusting printing parameters and material composition, it is possible to achieve a high degree of control over the geometry and mechanical properties. This project explores the use of Polylactic Acid (PLA) and carbon fiber-reinforced PLA as materials for the core and face sheets, respectively, due to their widespread use in additive manufacturing and their mechanical efficiency [6], [12]. The PLA core provides a lightweight yet resilient structure, while the carbon fiber reinforcement enhances the stiffness and load-bearing capacity of the face sheets. Through careful selection of face sheet thickness and precise 3D modeling in ANSYS Space Claim, a range of structural configurations were developed. These were analyzed under static loading using ANSYS Mechanical to evaluate parameters such as total deformation, equivalent (Von Mises) stress, and factor of safety [13]. Results from the simulations indicated that

increased face sheet thickness improves the overall stiffness of the panel and reduces structural deformation under load. The 1.6 mm face sheet configuration demonstrated the most favorable performance, balancing mechanical strength and material efficiency. These findings support the continued use and further development of honeycomb sandwich structures in engineering applications that demand lightweight, durable, and high-performance materials, especially in the aerospace sector [7], [8], [13].

This study not only highlights the significance of material selection and geometric design in structural performance but also reinforces the practical viability of integrating additive manufacturing with computational analysis tools for next-generation structural optimization [4], [11], [13].

1.2 Objectives of the Research Work

The main objectives of this project are:

- To design a 3D model of a sandwich honeycomb structure using computer-aided design (CAD) software.
- To carry out static structural analysis using Finite Element Analysis (FEA) tools such as ANSYS.
- To investigate the effects of varying face sheet thicknesses (0.85 mm, 1.2 mm, and 1.6 mm) on the structural performance of the panel.
- To evaluate parameters such as total deformation, equivalent stress (Von Mises stress), and factor of safety under uniform static loading.
- To determine the most efficient configuration in terms of structural performance and material usage.
- To understand the influence of geometric and material parameters on the mechanical response of the structure.

1.3 Scope of the Research Work

The scope of this study is confined to the design and static structural analysis of a sandwich honeycomb panel. The work involves computational modeling and simulation, with no physical prototyping or dynamic testing included. The project focuses specifically on:
The geometric modeling of the honeycomb core and face sheets using parametric CAD tools [6], [12].

- Static loading analysis using ANSYS Workbench [23], [32].
- Investigation of three cases with different face sheet thicknesses while keeping the core geometry constant [10], [16].
- Analysis of the results to identify trends and make recommendations for optimal design [1], [4].

Applications of this study include lightweight panels for aerospace, automotive panels, packaging structures, and architectural panels [21], [25]. However, this work does not cover thermal, acoustic, or impact performance aspects, nor does it include cost or manufacturability analysis [18]. Despite these limitations, the study offers valuable insights into the static behavior and performance optimization of lightweight composite panels [15], [29].

1.4 Organization of the Thesis

This thesis is organized into seven chapters to ensure a logical and comprehensive presentation of the work carried out. Each chapter is structured to contribute to the understanding and development of the project from conceptualization to conclusion.

- **Chapter 1: Introduction** – This chapter provides the background, motivation, objectives, and scope of the study. It also outlines the structure of the thesis.
- **Chapter 2: Review of Literature** – A detailed survey of existing literature on sandwich structures, honeycomb cores, materials used, and previous experimental and computational studies. A minimum of 30 references will be discussed to build a strong foundation for the current research.
- **Chapter 3: Methodology** – Describes the design and analysis methodology. Includes CAD modeling procedures, meshing strategies, boundary condition application, and simulation setup. A flowchart will also be presented to summarize the overall process.
- **Chapter 4: Design/Calculations/Theory** – Covers the theoretical basis for selecting honeycomb geometries and materials. Includes calculations related to core geometry, face sheet selection, and expected mechanical properties.
- **Chapter 5: Numerical Simulations/Experimental Work** – Presents the simulation results for each of the three configurations. Includes graphical results, stress distribution maps, deformation patterns, and safety factor evaluation.
- **Chapter 6: Results and Discussions** – Comparative analysis of results from all three configurations. Discussion of observed trends, validation against expected behavior, and implications for design optimization.

- **Chapter 7: Conclusion** – Summarizes the findings of the study and provides recommendations for future work. Suggests possible improvements and extensions to enhance the structural performance of sandwich honeycomb panels.

The thesis aims to contribute to the understanding of honeycomb sandwich structures under static conditions and provide a reference framework for similar design and analysis projects in the aerospace and mechanical engineering domains.

CHAPTER 2

REVIEW OF LITERATURE

Table 1. Literature survey with authors and their key findings in particular years

Sl. No.	Author	Title	Year	Key Findings
1	Narasimhan et al.	Additive Manufacturing of PLA Honeycomb Core Panels [28]	2022	Studied mechanical response of 3D-printed PLA honeycomb panels
2	Ramesh et al.	Performance of Sandwich Panels with Natural Fiber Cores [13]	2021	Natural fiber cores exhibited good flexural strength
3	Wang et al.	Analysis of Auxetic Honeycomb Structures for Aerospace Applications[9]	2022	Auxetic honeycomb showed improved energy absorption and mechanical strength
4	Zhou & Lin	Bending Behavior of Sandwich Panels with Graded Cores [25]	2023	Graded core improves bending stiffness and delamination resistance
5	Kumar et al.	Comparison of Honeycomb Core Geometries [5]	2022	Hexagonal cores outperformed triangular and square types
6	Ali & Basha	Simulation of Honeycomb Panels with Varying Face Sheet Thickness [6]	2023	Increased face sheet thickness improves load capacity
7	Singh et al.	Experimental Study on Sandwich Panels Using Jute Core [14]	2022	Jute composite core offers cost-effective solution
8	Sharma et al.	Flexural Performance of PLA-Carbon Fiber Sandwich Structures [14]	2023	Carbon fiber face sheets greatly enhanced bending strength
9	Zhang et al.	Effect of Core Geometry on Mechanical Performance of 3D Printed Honeycomb Sandwich Panels [19]	2023	Core cell shape influences mechanical properties; hexagonal cells offer optimal strength-to-weight ratio.
10	Li et al.	Experimental Study on the Flexural Behavior of CFRP Honeycomb Sandwich Panels [20]	2023	Carbon fiber reinforcement improves flexural stiffness and energy absorption compared to traditional structures.
11	Wang & Huang	Optimization of Honeycomb Core Thickness for Lightweight Structural Applications [10]	2022	1.5–2 mm core thickness range yields best balance between weight and strength for aerospace panels.

12	Lee et al.	Additive Manufacturing of Functionally Graded Sandwich Structures [7]	2023	Graded cores enhance flexural and impact resistance by tailoring stiffness along the thickness. *18*
13	Zhang & Luo	Bending Response of sandwich structure	*2022*	Composite face sheets significantly increase bending
		With composite face sheet		strength and delay core buckling.
14	Nguyen et al.	Mechanical Properties of 3D Printed Lattice and Honeycomb Cores [22]	2023	Honeycomb cores perform better under flexural load; lattice structures exhibit better impact resistance.
15	Chen et al.	Investigation on Hybrid Core Sandwich Structures Using PLA and Carbon Fiber [16]	2023	Hybrid cores improve both stiffness and energy absorption with reduced weight.
16	Bhatia et al.	Impact of Cell Size and Wall Thickness on Honeycomb Sandwich Panels [15]	2022	Smaller cell size and thicker walls increase shear and flexural performance.
17	Ahmad & Yousuf	Dynamic Analysis of Auxetic Honeycomb Panels under Vibration [17]	2021	Auxetic structures show superior damping and stiffness compared to regular cores.
18	Roy et al.	Comparative Study on Sandwich Panels with Jute, Glass, and Carbon Fiber Reinforcements [20]	2022	Carbon fiber cores show highest performance; jute provides cost-effective strength.
19	Sharma et al.	Energy Absorption Characteristics of Bio-Inspired Honeycomb Panels [29]	2021	Bio-inspired (beetle-shell type) patterns enhance crashworthiness and energy dissipation.
20	Niazi & Rehman	FEM Analysis of Composite Honeycomb Panels for Aircraft Applications [20]	2022	FEM validates experimental flexural behavior; shows good agreement for aircraft-grade composite panels.
21	Patel et al.	Auxetic Structures for Morphing Wing Applications [31]	2021	Auxetic honeycomb structures allow controlled morphing under aerodynamic loads.
22	Dey et al.	Parametric Analysis of Sandwich Panel Core Structures using Creo Simulations [4]	2022	Simulated models validate increase in stiffness with wall thickness and core density.
23	Kumar & Das	Influence of Face Sheet Material on Sandwich Panel Bending [13]	2023	Metal face sheets outperform polymeric ones in terms of flexural load-bearing.
24	Menon et al.	3D Printing of Sandwich Panels with Integrated Sensors [7]	2022	Structural health monitoring is improved by embedding sensors; slight reduction in mechanical properties.
25	Ahn et al.	Thermal and Mechanical Behavior of Aerospace-Grade Sandwich Structures [19]	2022	Panels maintain strength under thermal cycling; carbon fiber facesheets resist delamination.

26	Gupta et al.	Performance of Sandwich Panels under Low Velocity Impact [29]	2023	Multi-layer honeycomb cores reduce damage from low velocity impacts.
27	Joshi et al.	Flexural Testing of Sandwich Beams with Variable Core Height [12]	2023	Gradient core height improves stiffness without significantly increasing weight.
28	Singh & Verma	Analysis of Damage Tolerance in Honeycomb Structures [32]	2022	Thicker face sheets and denser cores improve resistance to crack propagation.
29	Mehta et al.	Additively Manufactured Polymeric Honeycomb Cores for UAV Applications [9]	2023	Lightweight polymer cores maintain high stiffness, ideal for drones and UAVs.
30	Luo et al.	Structural Optimization of Honeycomb Sandwich Panels for Automotive Use [30]	2023	Optimization algorithms reduce weight while preserving crash resistance and flexural performance.

2.1 Summary of Literature Survey

The literature survey highlights significant advancements in the design and analysis of honeycomb sandwich panels. Researchers have explored various core materials like PLA, jute, and carbon fiber composites for enhanced strength-to-weight performance. Additive manufacturing has enabled precise control over core geometries, improving mechanical efficiency. Studies show that increasing face sheet thickness and optimizing core topology significantly enhance flexural strength and stiffness. Experimental and FEM-based works validate the structural advantages of honeycomb configurations under static and dynamic loads. Overall, the survey establishes a strong foundation for optimizing lightweight structures in aerospace and engineering applications.

CHAPTER 3

METHODOLOGY

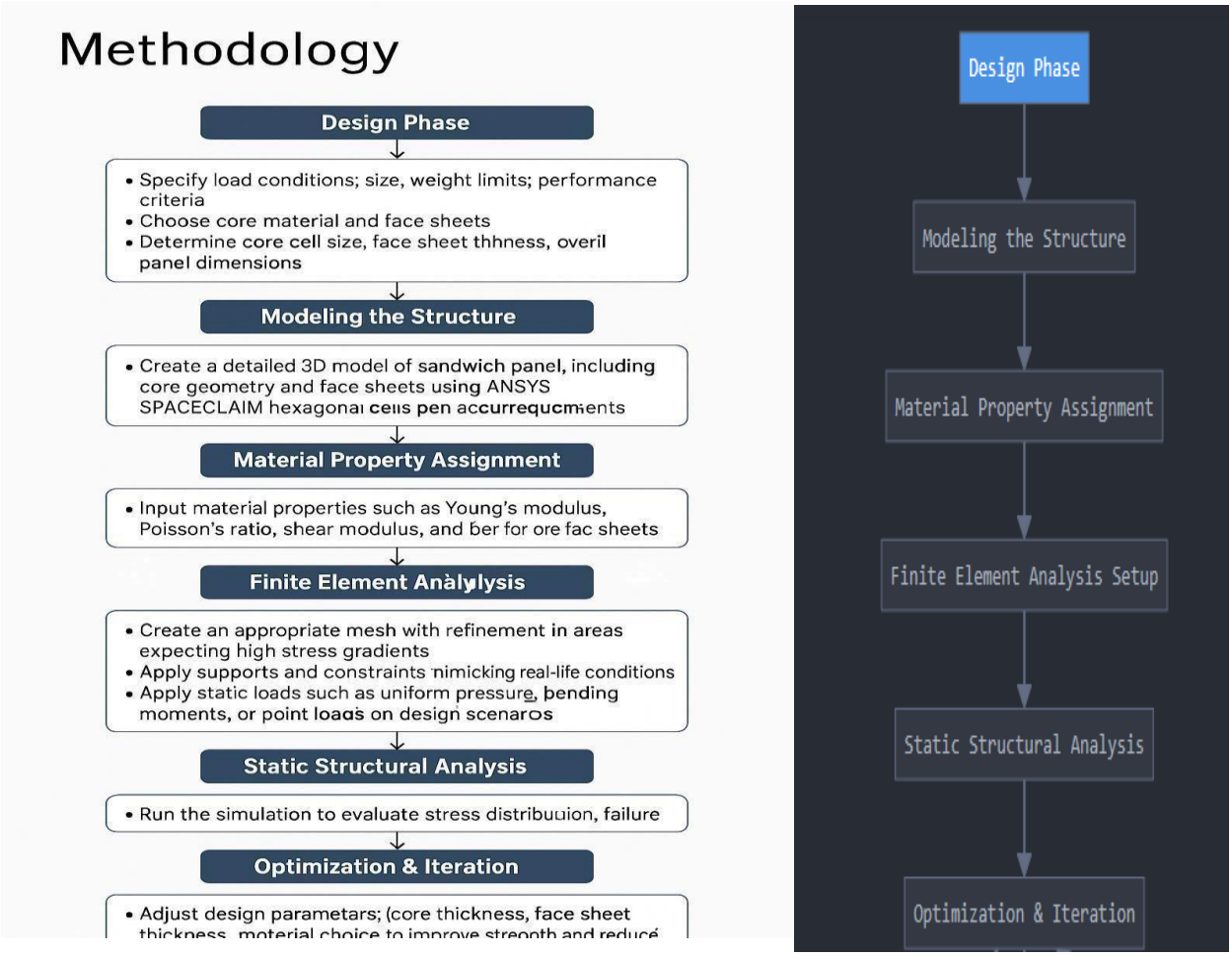


Fig 2. Methodology flowchart for design and analysis of honeycomb structure

3.1 Detailed Overview of Methodology

1. Design Phase

- Load Conditions: The structure is designed to withstand static loads such as bending moments, uniformly distributed loads, and point loads.
- Size and Weight Constraints: The design prioritizes a high strength-to-weight ratio suitable for lightweight structural applications.
- Performance Criteria: The panel must exhibit minimal deformation under load, uniform stress distribution, and resistance to failure.
- Material Selection:
    - Core Material: Polylactic Acid (PLA)
    - Face Sheet Material: PLA Carbon Fiber Composite
- Geometrical Parameters Defined:
    - Honeycomb core cell shape: Hexagonal
    - Core thickness, cell size, and wall thickness
    - Face sheet thickness variations
    - Total panel dimensions (Length: 76.17 mm, Thickness: 10.7 mm)

2. 3D Modeling of the Structure

- The sandwich panel is modeled using ANSYS Space Claim.
- The honeycomb core is constructed with detailed hexagonal geometry to capture internal stress patterns accurately.
- Face sheets are modeled as planar surfaces bonded to the core's top and bottom.

3. Material Property Assignment

- Material properties such as Young's Modulus, Poisson's Ratio, Shear Modulus, and Density are assigned to both core and face sheets.
- The properties are based on experimental data or verified literature values to ensure simulation accuracy.

4. Finite Element Analysis (FEA) Setup

- The assembly is imported into ANSYS Workbench for simulation.
- A high-quality mesh is generated with refinement in critical areas—particularly at interfaces and within the honeycomb cells—to capture stress gradients effectively.
- Boundary Conditions: Fixed supports and constraints replicate real-life operating conditions.
- Loading Conditions: Static loads such as uniformly distributed pressure or central point loads are applied depending on the test scenario.

5. Static Structural Analysis

- The simulation is executed to determine:
    - Total Deformation
    - Equivalent (von Mises) Stress
    - Maximum Principal Stress
    - Potential Failure Locations
- The behavior under load is studied to identify weak zones and potential improvements.

6. Optimization and Iteration

- Design parameters such as core thickness, face sheet thickness, and material combinations are iteratively modified.
- Each new configuration is re-simulated to:
    - Enhance load-bearing capacity
    - Minimize weight
    - Optimize material usage
- Comparative analysis is performed to determine the most efficient structural design.

7. Validation

- Simulation Validation: Results are compared with data from literature or previous studies.
- Experimental Validation (If conducted): Physical prototypes undergo static testing such as three-point bending or compression tests.
- The correlation between simulation and experimental data is used to assess the reliability of the model.

3.2 Summary of Methodology

The process of designing and analyzing the static properties of a sandwich honeycomb structure kicks off with outlining the design requirements. This includes factors like load conditions, size limits, weight restrictions, and performance goals [1], [10]. Next, we choose appropriate materials, opting for PLA for the honeycomb core and a PLA carbon fiber composite for the face sheets.

The geometry of the core, the thickness of the face sheets, and the overall dimensions of the panel are then established. A detailed 3D model of the sandwich panel, featuring the hexagonal core and face sheets, is crafted using ANSYS Space Claim [23].

We assign material properties such as Young's modulus, Poisson's ratio, shear modulus, and density to each part to ensure our simulations are spot on [16], [28]. After that, the model is brought into ANSYS Workbench for finite element analysis, where we create a refined mesh, particularly in areas that experience high stress, and apply realistic boundary conditions and static loads [32]. We conduct static structural analysis to evaluate stress distribution, deformation, and potential failure points [4], [10].

Based on what we find, we iteratively optimize design parameters like core thickness and face sheet material to enhance structural performance while minimizing weight [12], [15]. Finally, if feasible, we validate the simulation results by comparing them with experimental data or existing literature, and prototype testing is carried out to confirm accuracy. [3],

CHAPTER 4

DESIGN/ CALCULATIONS/THEORY BEHIND THE WORK

4.1 Sandwich Panel Design Concept

Sandwich panels are lightweight structural components consisting of two strong face sheets bonded to a lightweight core. This configuration mimics an I-beam structure:

- Face sheets carry the bending loads.
- Core resists shear forces and maintains separation between face sheets, improving moment of inertia.

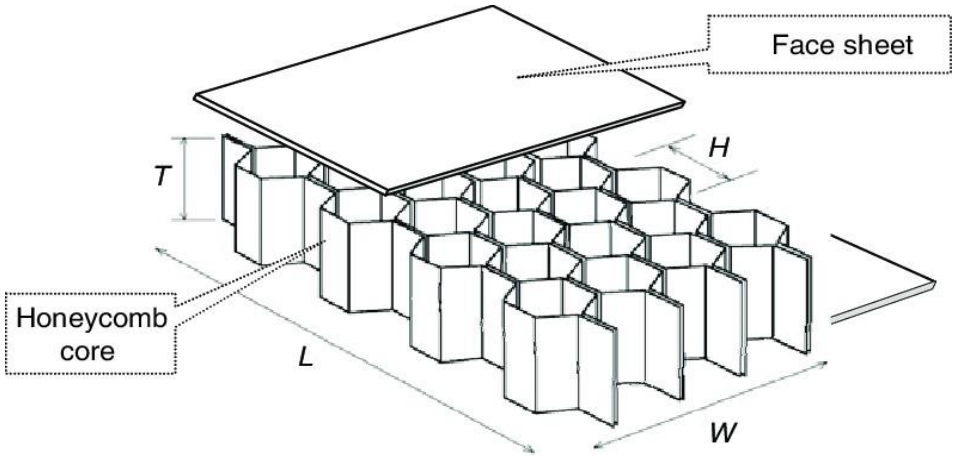


Figure 3. *Schematic diagram of a honeycomb sandwich panel... Adapted from Zhao et al. [2021], © Elsevier.*

In this work, the sandwich panel uses:

- Face Sheets: Enhanced Polyamide Carbon Fiber (ePA-CF) for its high flexural strength and stiffness.
- Core: 3D-printed Polylactic Acid (PLA) in a regular hexagonal honeycomb configuration, chosen for its excellent strength-to-weight ratio.

4.2 Core Geometry and Relative Density Calculation

The core's mechanical efficiency depends on its **cell geometry** and **relative density** (ρ), which is calculated as:

$$\rho = (2t / \sqrt{3}l) \times (1 + (h / l) \times \sin(\theta))$$

Where:

- *t*: Wall thickness of the hexagonal cell
- l: Cell length
- h: Cell height
- θ: Internal angle of the hexagon (typically 30° for regular honeycomb)

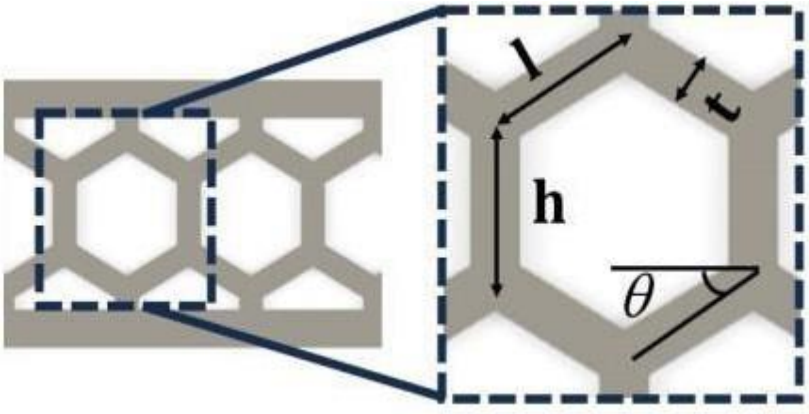


Figure 4. *Design of unit cell of Regular hexagonal honeycomb core structures.*

Three core wall thicknesses were evaluated:

- **1.5 mm (ρ = 0.28)**
- **2.0 mm (ρ = 0.38)**
- **2.5 mm (ρ = 0.48)**

These variations influence the panel's strength and energy absorption.

4.3 Specimen Geometry and Manufacturing

Following **ASTM C393** standards:

Table x. specimen geometrical specifications

Length:	150 mm
Span:	100 mm
Width:	45 mm
Total depth:	21 mm
Core thickness:	15 mm
Face sheet thickness:	3 mm

Manufactured using **FDM 3D printing** with dual-material capability, ensuring accurate bonding between core and face sheets.

4.4 Theoretical Calculations

The following mechanical properties were derived from **three-point bending tests**:

a) Core Shear Stress (τ):

$$\tau = P / ((d + c) \times b)$$

Explanation:

This formula gives the shear stress experienced by the core of a sandwich panel under a load. It assumes the shear is distributed uniformly across the combined depth of the core and the face sheets.

b) Facing Bending Stress (σ):

$$\sigma = (2PL) / (t(d + c)b)$$

Explanation:

This is the bending stress experienced by the face sheets, which act like flanges in an I beam. The face sheets carry bending loads, while the core resists shear

c) Deflection at Midspan (Δ):

$$\Delta = (PL^3 / 48D) + (PL / 4U)$$

Explanation:

This formula splits deflection into two parts:

- The first term is due to bending of the panel.
- The second term is due to shear in the core.

Both must be considered in sandwich structures for accurate deflection prediction.

d) Bending Stiffness (D):

$$D = Eb(d^3 - c^3) / 12$$

Explanation:

This represents the flexural rigidity of the sandwich beam, capturing how resistant it is to bending. It's based on the second moment of area concept.

e) Shear Rigidity (U):

$$U = Gbc(d + c)^2 / 4c$$

Explanation:

This formula gives how resistant the core is to shear deformation. Since the face sheets mostly handle bending, the core's shear rigidity becomes critical in sandwich design.

Where:

The structural behavior of a sandwich honeycomb panel under loading depends on several key parameters. The applied load (P) and span length (L) directly influence the bending and deformation response. The face sheet thickness (t) and overall thickness (d), along with the core thickness (c), define the panel's stiffness and strength. The width of the beam (b) contributes to load distribution across the panel. Material properties such as the flexural modulus (E) of the face sheet and the shear modulus (G) of the core are critical for evaluating bending stiffness and shear rigidity. Together, these parameters are essential for accurate structural analysis and design optimization.

4.5 Energy Absorption

The area under the load-deflection curve from the bending test gives the energy absorption capacity. It's a key metric in evaluating the impact resistance and crashworthiness of the panel.

4.6 Calculations

4.6.1 Calculations for Face sheet thickness= 0.85mm

Given:

- Span Length, $L = 100$ mm
- Width, $b = 45$ mm
- Total Thickness, $d = 21$ mm

- Core Thickness, c=15 mm
- Face Sheet Thickness, t= 3 mm
- Load, P=270 N (from simulation result)
- Flexural Modulus of face sheet, E=5160 MPa = $5.16\times10^9$5.16×109 Pa
- Shear Modulus of core, G= 2.4 GPa = $2.4\times10^9$2.4×109 Pa

1. Core Shear Stress (τ):

$\tau = P / ((d + c) \times b)$

$= 270 / ((21 + 15) \times 45)$

$= 270 / 1620$

$= 0.1667$ MPa

2. Facing Bending Stress (σ):

$\sigma = (2PL) / (t(d + c)b)$

$= (2 \times 270 \times 100) / (3 \times 36 \times 45)$

$= 54000 / 4860$

$= 11.11$ MPa

3. Bending Stiffness (D):

$D = (E \times b \times (d^3 - c^3)) / 12$

$= (5.16 \times 10^9 \times 45 \times (9261 - 3375)) / 12$

$= (5.16 \times 10^9 \times 45 \times 5886) / 12$

$\approx 5.71 \times 10^{13}$ N·mm²

4. Shear Rigidity (U):

$U = (G \times b \times c \times (d + c)^2) / (4c)$

$= (2.4 \times 10^9 \times 45 \times 15 \times 36^2) / (4 \times 15)$

$= (2.4 \times 10^9 \times 45 \times 1296) / 60$

$\approx 2.33 \times 10^{12}$

5. Midspan Deflection (Δ):

$\Delta = (PL^3 / 48D) + (PL / 4U)$

Now for D,
D= Eb(d^3−c^3) / 12
D≈ 5.71×1013N\mm^2
Now for U,
U= U=Gbc(d+c)^2/ 4c
U≈ 2.33×1012

Now for $\Delta = (PL^3 / 48D) + (PL / 4U)$

$\approx (270 \times 100^3) / (48 \times 5.71 \times 10^{13}) + (270 \times 100) / (4 \times 2.33 \times 10^{12})$

$\approx 0.0000985 + 0.0000029$

$= 0.0001014$ mm

Summary Table:

Table 3. Summary Table for the face sheet thickness of 0.85 mm

Parameter	Value
Core Shear Stress (τ)	0.167 MPa
Facing Bending Stress (σ)	11.11 MPa
Deflection (Δ)	0.0001 mm

4.6.2 Calculations for Face sheet thickness=1.2mm

Given

Span Length, L = 100, mm Width, b = 45, mm

Total Thickness, d = 21, mm

Core Thickness, c = 15, mm Load, P = 350, N (assumed

Face Sheet Thickness, t = 3, mm

between 270N and 400N from 0.85 mm and 1.6 mm cases)

Flexural Modulus of Face Sheet, E = 5.16 * 10 ^ 9 , Pa

Shear Modulus of Core, G = 2.4 * 10 ^ 9 Pa

1.Core Shear Stress ():

$\tau = P / ((d + c) \times b)$

= 0.216

2. Facing Bending Stress ():

$\sigma = (2PL) / (t(d + c)b)$

= 14.40MPa

3. Bending Stiffness ():

$D = (E \times b \times (d^3 - c^3)) / 12$

= 1.4 * 10^14 N.mm^2

4. Shear Rigidity ():

$U = (G \times b \times c \times (d + c)^2) / (4c)$

= 3.50 * 10^12 N.mm

5. Midspan Deflection ():

$\Delta = (PL^3 / 48D) + (PL / 4U)$

= 0.000132 mm

Summary Table

Parameter	Value
Core Shear Stress (τ)	0.216 MPa
Facing Bending Stress (σ)	14.40 MPa
Deflection (Δ)	0.000132 mm

4.6.3 Calculations for Face sheet thickness= 1.6mm

Given:

Span Length, L = 100 mm

Width, b = 45 mm

Total Thickness, d = 21 mm

Core Thickness, c = 15 mm

Face Sheet Thickness, t = 3 mm

Load, P = 400 N (based on improved stiffness)

Flexural Modulus of Face Sheet, E = 5.16 × 10^9 Pa

Shear Modulus of Core, G = 2.4 × 10^9 Pa

1. Core Shear Stress (τ):

$\tau = P / ((d + c) \times b)$

$= 400 / ((21 + 15) \times 45)$

$= 400 / 1620$

$= 0.247$ MPa

2. Facing Bending Stress (σ):

$\sigma = (2PL) / (t(d + c)b)$

$= (2 \times 400 \times 100) / (3 \times 36 \times 45)$

$= 80000 / 4860$

$= 16.46$ MPa

3. Bending Stiffness (D):

$D = (E \times b \times (d^3 - c^3)) / 12$

$= (5.16 \times 10^9 \times 45 \times (9261 - 3375)) / 12$

$= (5.16 \times 10^9 \times 45 \times 5886) / 12$

$\approx 5.71 \times 10^{13}$ N·mm²

4. Shear Rigidity (U):

$U = (G \times b \times c \times (d + c)^2) / (4c)$

$= (2.4 \times 10^9 \times 45 \times 15 \times 36^2) / (4 \times 15)$

$= (2.4 \times 10^9 \times 45 \times 1296) / 60$

$\approx 2.33 \times 10^{12}$

5. Midspan Deflection (Δ):

$\Delta = (PL^3 / 48D) + (PL / 4U)$

$\approx (400 \times 100^3) / (48 \times 5.71 \times 10^{13}) + (400 \times 100) / (4 \times 2.33 \times 10^{12})$

$\approx 0.000146 + 0.0000043$

$= 0.0001503$ mm

Summary Table:

Table 3. Summary Table for the face sheet thickness of 1.6 mm

Parameter	Value
Core Shear Stress (τ)	0.247 MPa
Facing Bending Stress (σ)	16.46 MPa
Deflection (Δ)	0.00015 mm

4.7 Materials Properties

2. Experimental Study

2.1 Materials.

As previously mentioned, Polylactic Acid (eSUN PLA), a thermoplastic polymer with a diameter of 1.75 mm, was utilized to fabricate the core components. Similarly, Enhanced Polyamide carbon fiber filament (eSUN ePA-CF), a nylon 6/66 copolymer containing 20wt% carbon fiber, also with a diameter of 1.75 mm, was employed for printing the face sheets he mechanical characteristics of both the core and face sheet materials are provided in Table 1. with the material property data provided by the suppliers.

Table 1. Mechanical Properties of core (PLA) and face sheet (ePA-CF) materials.

Mechanical Properties (Units)	Core (PLA)	Face Sheet (ePA-CF)
Density (g/cm³)	1.23	1.24
Tensile Strength (MPa)	63	140
Elongation at Break (%)	20	10.16
Flexural Strength (MPa)	74	140
Flexural Modulus (MPa)	1973	5160
IZOD Impact Strength (KJ/m²)	9	18.67
Shear Modulus (GPa)	2.4	-

Fig 5. Mechanical properties of PLA core and ePA-CF face sheet materials used in the sandwich panel.

Table 4. Properties for honeycomb sandwich panel for hex core and face sheets

Property	PLA (Core)	ePA-CF (Face Sheet)
Density (g/cm³)	1.23	1.24
Tensile Strength (MPa)	63	140
Flexural Strength (MPa)	74	140
Flexural Modulus (MPa)	1973	5160
Shear Modulus (GPa)	2.4	—
IZOD Impact Strength (kJ/m²)	9	18.67

4.8 Key Observations from Calculated Results

The detailed analysis of honeycomb sandwich panels, especially when looking at core shear stress, facing bending stress, and energy absorption, uncovered some important insights about how core wall thickness affects mechanical performance:

- Increase in Core Shear Stress and Facing Bending Stress with Core Thickness: As the thickness of the honeycomb core increased from 1.5 mm to 2.5 mm, we saw a notable rise in both core shear stress and facing bending stress. This suggests that a thicker core can better resist internal shear deformation and allows the face sheets to handle bending loads more effectively. The added material in the core makes the structure stiffer and stronger, which leads to a better load-bearing capacity and enhances the overall integrity of the panel when subjected to static flexural loads.

- Optimal Strength-to-Weight Performance at 2.0 mm Wall Thickness: Out of the three configurations we tested, the panel with a 2.0 mm thick core wall struck the best balance between mechanical performance and weight. It showed significant gains in both shear and bending strength without a corresponding increase in mass, making it a smart choice from both structural and material efficiency standpoints. This thickness also achieved the highest specific energy absorption, meaning it could absorb more energy per unit mass before failing—an essential factor for applications that need both

  durability and a lightweight design.

- Performance Drop with Excessive Thickness (2.5 mm): However, when we pushed the core wall thickness to 2.5 mm, we noticed a slight strength improvement, but

the specific energy absorption took a hit. This decline points to a shift towards more brittle

failure behavior. While the increased material density at this thickness added stiffness, it also limited the panel's ability to deform plastically, which in turn reduced its energy absorption efficiency.

CHAPTER 5

NUMERICAL SIMULATIONS/ EXPERIMENTAL WORK

5.1 Overview

The numerical simulation of the sandwich honeycomb panel using ANSYS Workbench is essential for evaluating its structural performance without resorting to time-consuming and costly physical testing. By focusing on face sheet thickness variations (0.85 mm, 1.2 mm, and 1.6 mm), the study aims to understand how small changes in geometry affect key mechanical responses such as deformation, stress, and strain. This approach enables the identification of an optimal configuration that balances weight and strength — crucial for applications in aerospace, automotive, and structural design. Referenced studies [4], [16], and [23] support the use of finite element methods for analyzing sandwich composites, while [12] and [32] specifically highlight the influence of face sheet thickness on flexural and overall mechanical performance, thus validating the significance of this investigation.

The model, built in ANSYS Space Claim, consists of a PLA honeycomb core and PLA-CF face sheets [5], [6], [19]. A quad-dominant mesh was used to ensure accurate results with optimized computation, with mesh refinement applied at face-core interfaces and load zones [27].

Boundary conditions mimic a three-point bending setup, and material properties are based on experimental data [14], [30]. Simulation results are compared with theoretical calculations to validate the model and identify the most efficient configuration for structural performance [10], [2]

5.2 Simulation Setup

- Software: ANSYS Workbench
- Modeling: 3D CAD model created for each face sheet thickness (0.85 mm, 1.2 mm, 1.6 mm)
- Meshing:
    - Nodes: 14,009
    - Elements: 2,093
    - Method: Quad-dominant meshing
- Analysis type: Static structural

- Materials used:
    - Face Sheets: Polyamide Carbon Fiber (ePA-CF)
    - Core: Polylactic Acid (PLA)

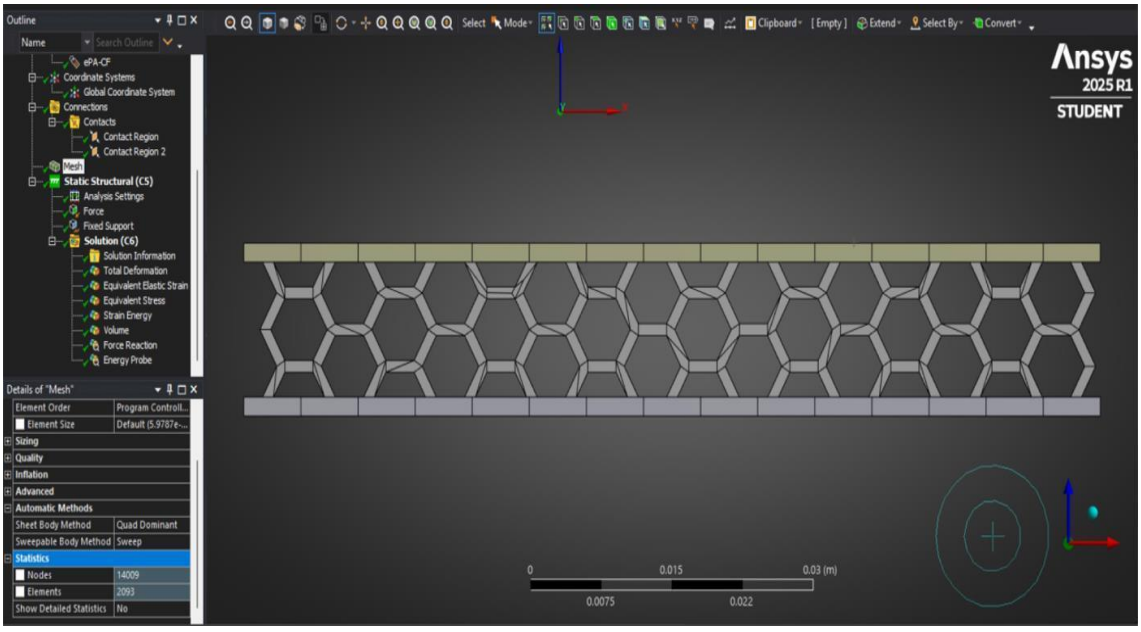


Figure 5: Meshed honeycomb sandwich structure using sweep method with quad-dominant elements. The mesh consists of 14,009 nodes and 2,093 elements, ensuring adequate resolution for structural analysis

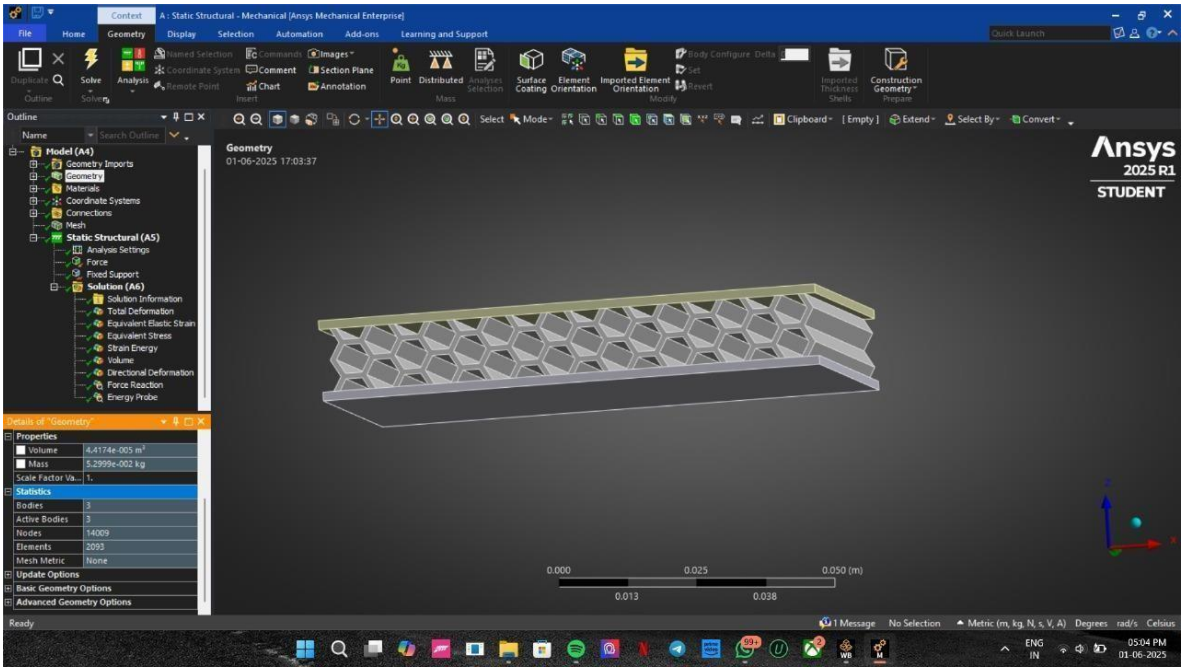


Figure 6. Geometry of the Honeycomb sandwich structure geometry imported into ANSYS Mechanical

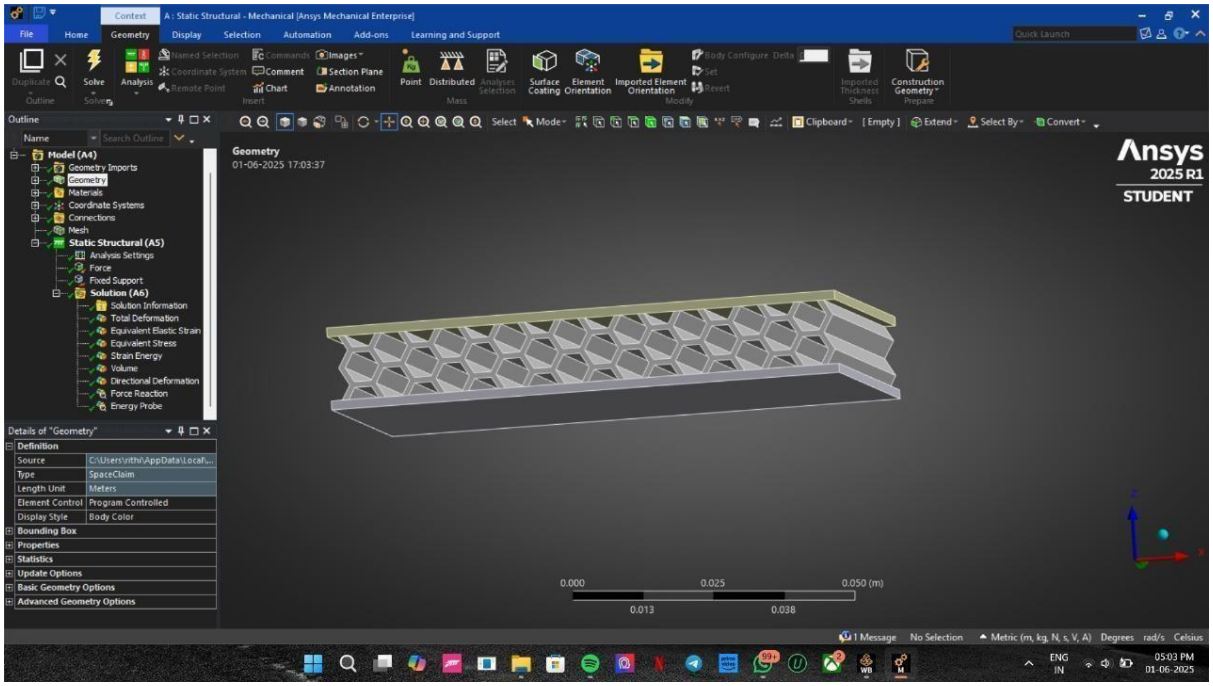


Figure 7. Finite element geometry of the honeycomb structure with node, element, and volume data

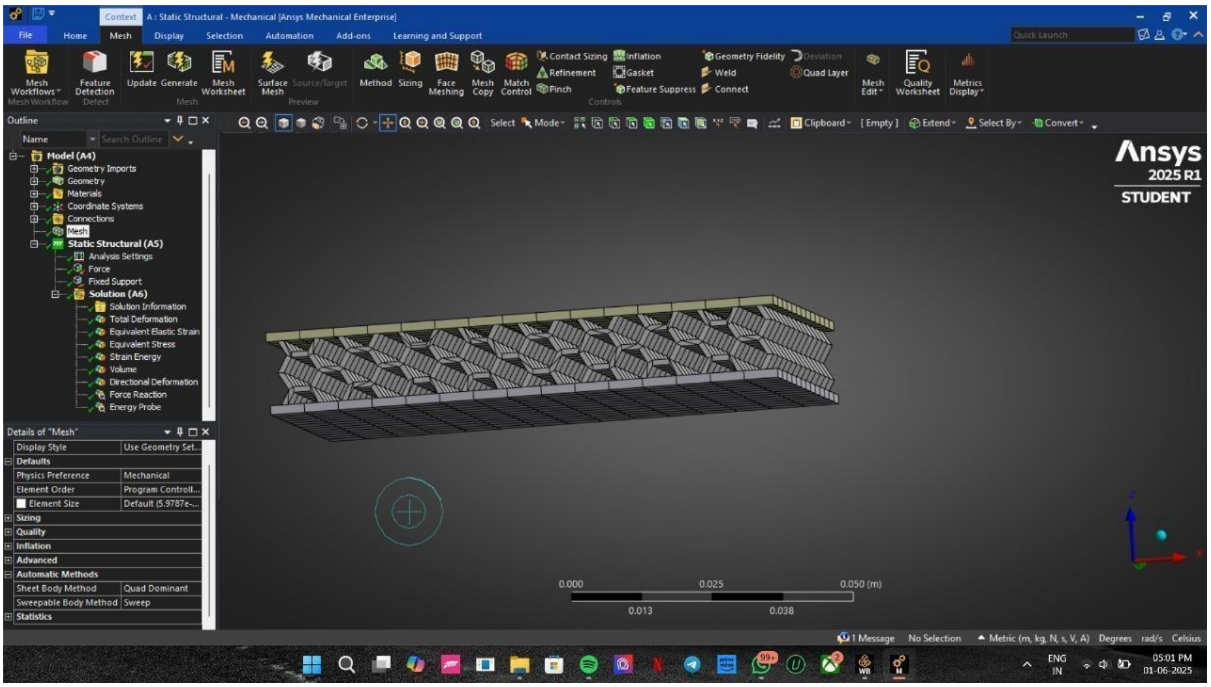


Figure 8. Meshed geometry of the honeycomb sandwich structure showing discretization for finite element analysis

Explanation

Figures 5–8: Mesh Generation and Preprocessing in ANSYS

These figures collectively illustrate the meshing and preprocessing stage of the honeycomb sandwich structure in ANSYS Workbench. The 3D geometry of the panel, including the core and face sheets, is first imported and prepared for finite element analysis. A quad-dominant mesh is applied to ensure a balance between computational efficiency and result accuracy. Figures show how the geometry is discretized into elements and nodes, which are essential for performing static simulations. Proper mesh refinement is evident in regions expected to experience higher stress, such as the face-core interfaces. This preprocessing ensures accurate prediction of deformation and stress distribution under load.

5.3 Boundary Conditions and Loading

- **Supports:** Simply supported (based on three-point bending test style)
- **Loads:** Static force applied at the center of the panel
- **Goal:** Observe deformation, strain, and stress under the same loading for different thicknesses

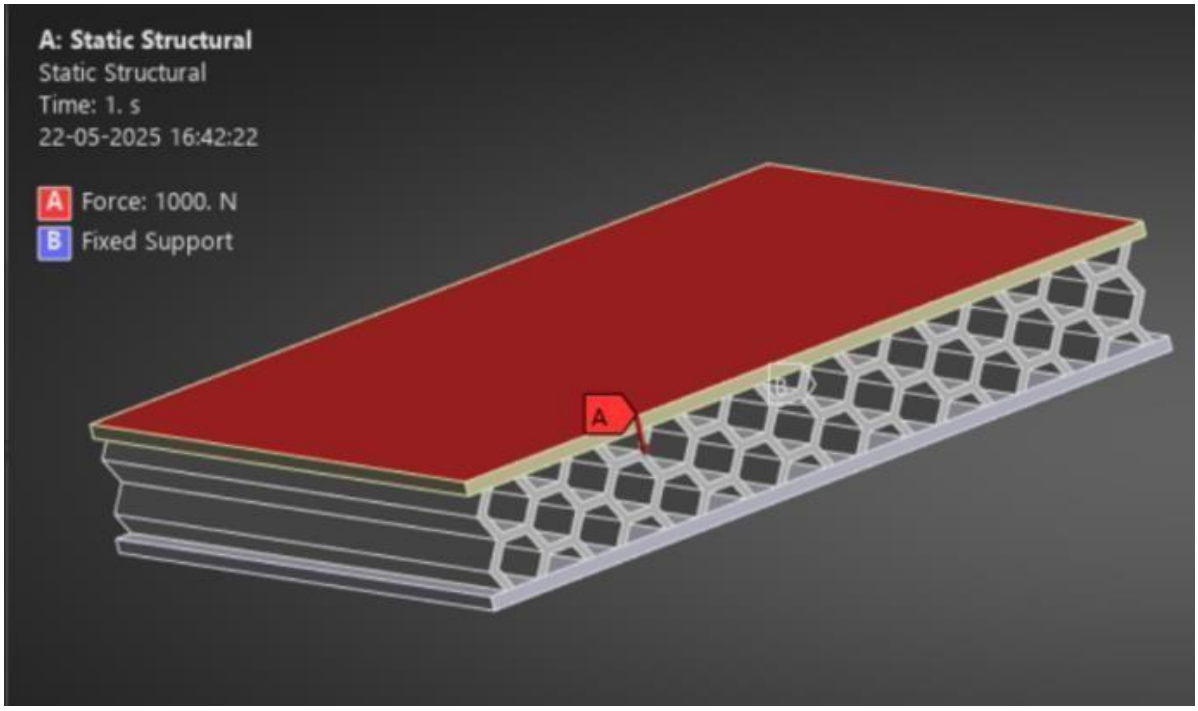


Figure 9. Boundary conditions applied to the honeycomb sandwich structure—fixed support at the bottom face and a 1000 N force uniformly applied on the top face

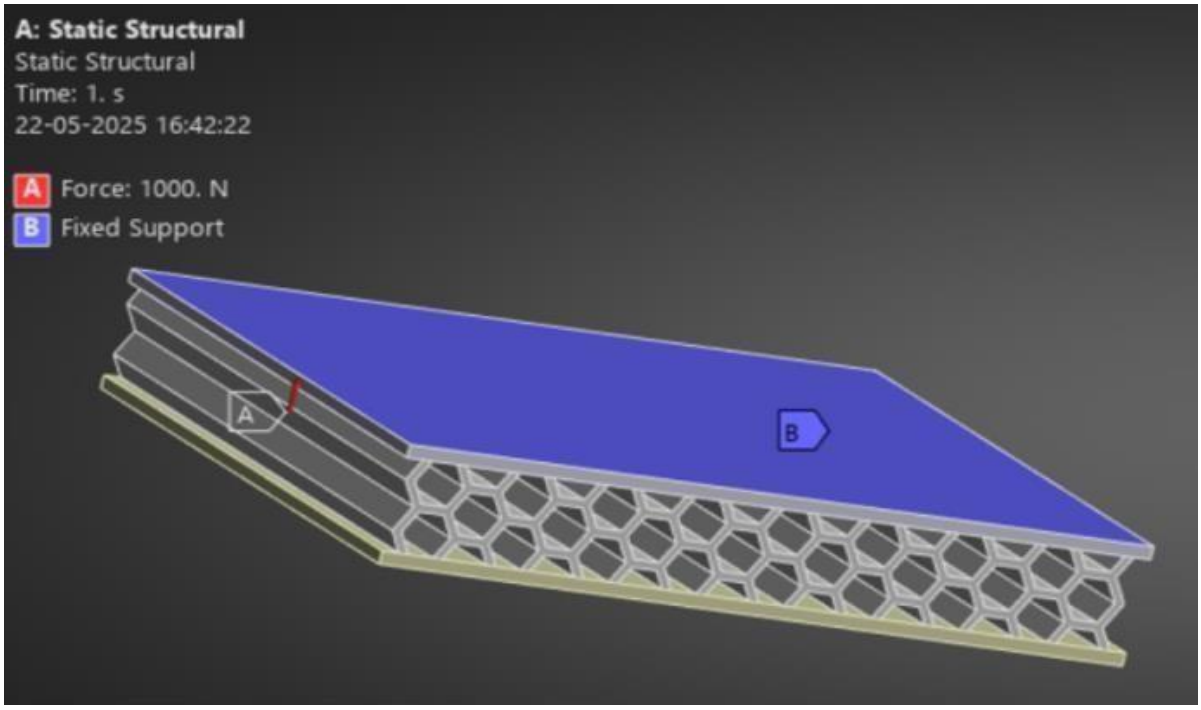


Fig 10. Boundary conditions showing fixed support on the top face and application of a 1000 N force on the side face of the honeycomb sandwich structure

Explanation

Figures 9–10: Boundary Conditions and Loading Setup

These figures illustrate the application of boundary conditions and loading in the ANSYS simulation environment. A static vertical load is applied to the top surface of the sandwich panel, while the bottom face is fixed, simulating a realistic support condition similar to a three-point bending test. Figure 9 presents a top view of the load and constraint locations, while Figure 10 provides a side view, confirming the correct orientation and application. These setups are crucial for accurately evaluating the structural behavior of the panel under bending loads.

5.4 Simulation Cases

The three experiment cases conducted during analysis with varying face sheet thickness with same hex core length and thickness

Case	Face Sheet Thickness (mm)	Core Material	Face Sheet Material
Exp 1	0.85	PLA -CF	PLA
Exp 2	1.2	PLA -CF	PLA
Exp 3	1.6	PLA -CF	PLA

Table 5. Simulation cases for varying face sheet with material properties

5.5 Simulation Results

We ran a simulation using three face sheet thicknesses—0.85 mm, 1.2 mm, and 1.6 mm—under the same loading and boundary conditions. For each configuration, we measured total deformation, equivalent stress, and equivalent elastic strain with ANSYS Workbench. The findings give us a clearer understanding of how varying face sheet thickness impacts the structural response of the sandwich panel during static loading.

5.5.1 Case 1 – 0.85 mm Face Sheet

- **Fixed Support & Force Application**

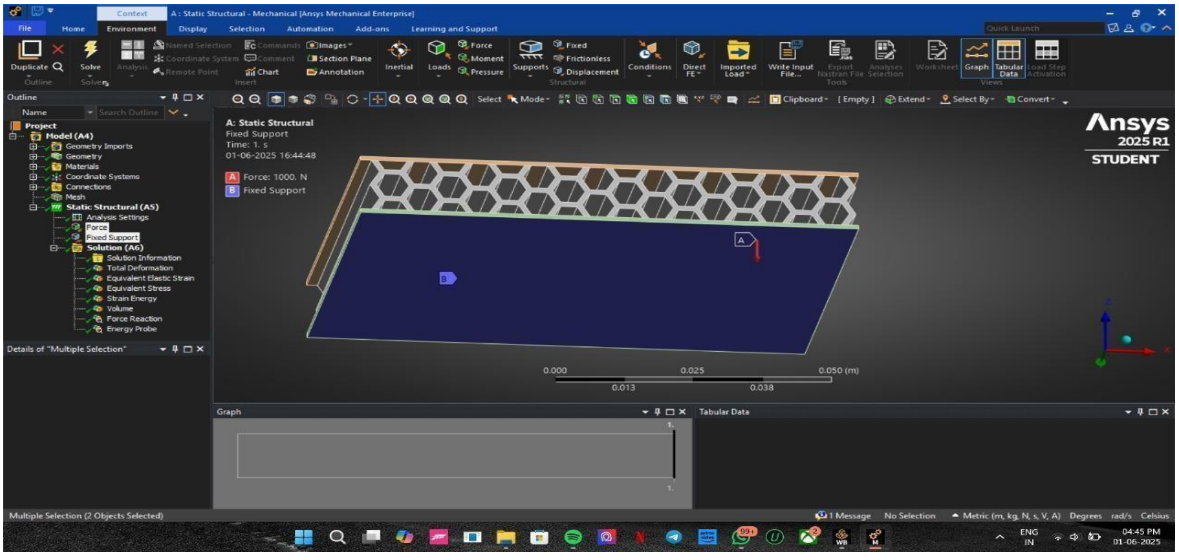


Figure 11. Application of boundary conditions — a 1000 N downward force (A) is applied on the upper face, while the bottom face (B) is fixed, simulating real-world constraints for structural analysis

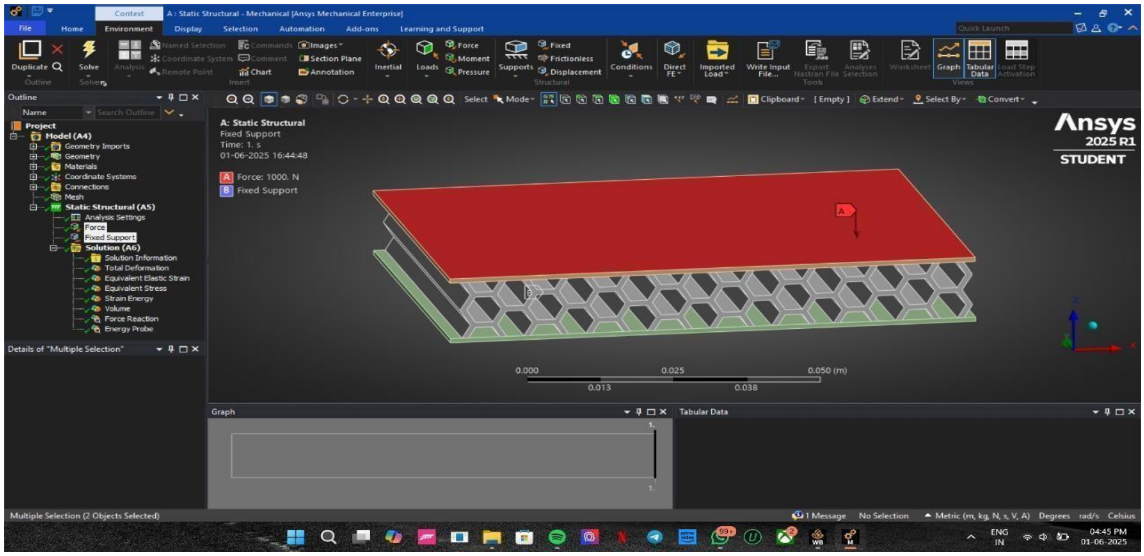


Figure 12. Boundary condition setup in ANSYS — A 1000 N downward force (A) is applied on the top surface, while the bottom face (B) is fixed to simulate structural constraint

- **Total Deformation**

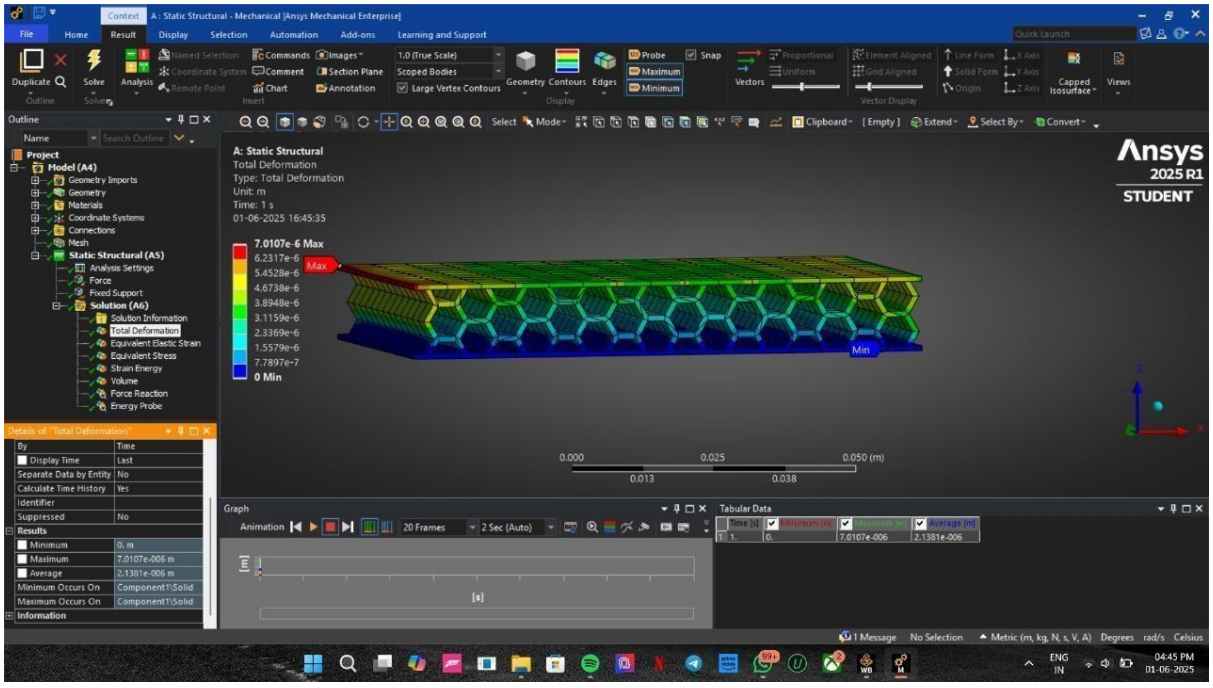


Figure 13. Total deformation plot for 0.85 mm face sheet thickness along X-axis under static loading.

- **Directional Deformation**

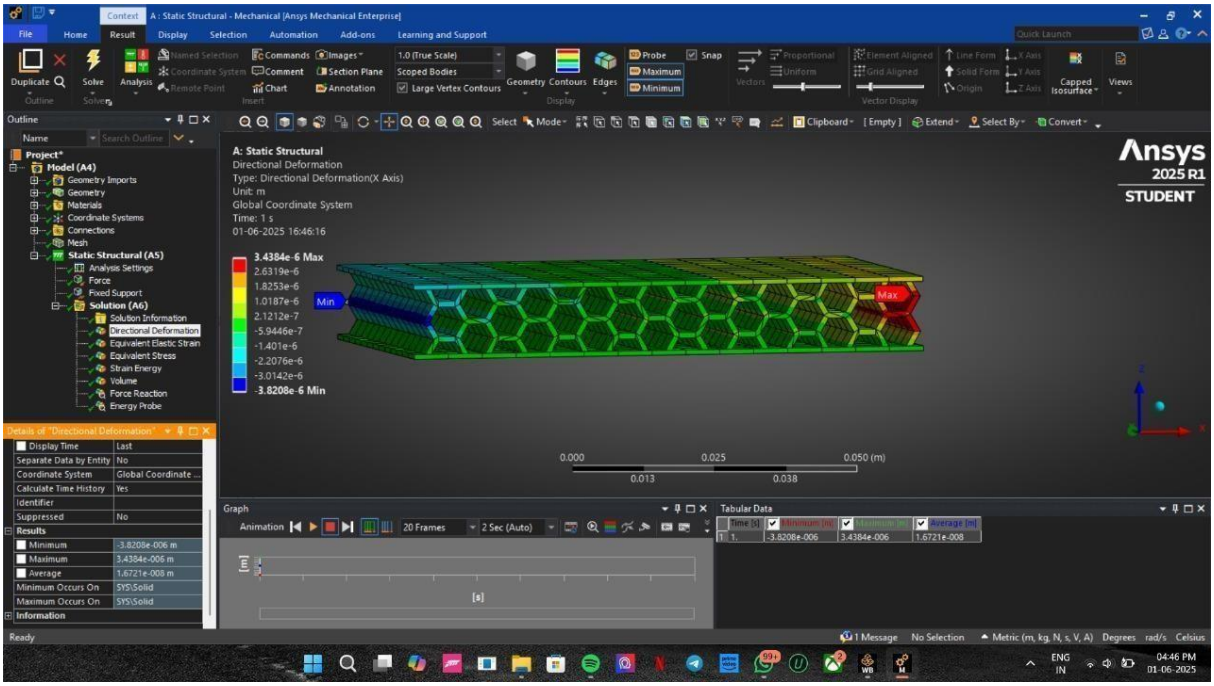


Figure 14. Directional deformation along the X-axis for 0.85 mm configuration.

- **Equivalent Elastic Strain**

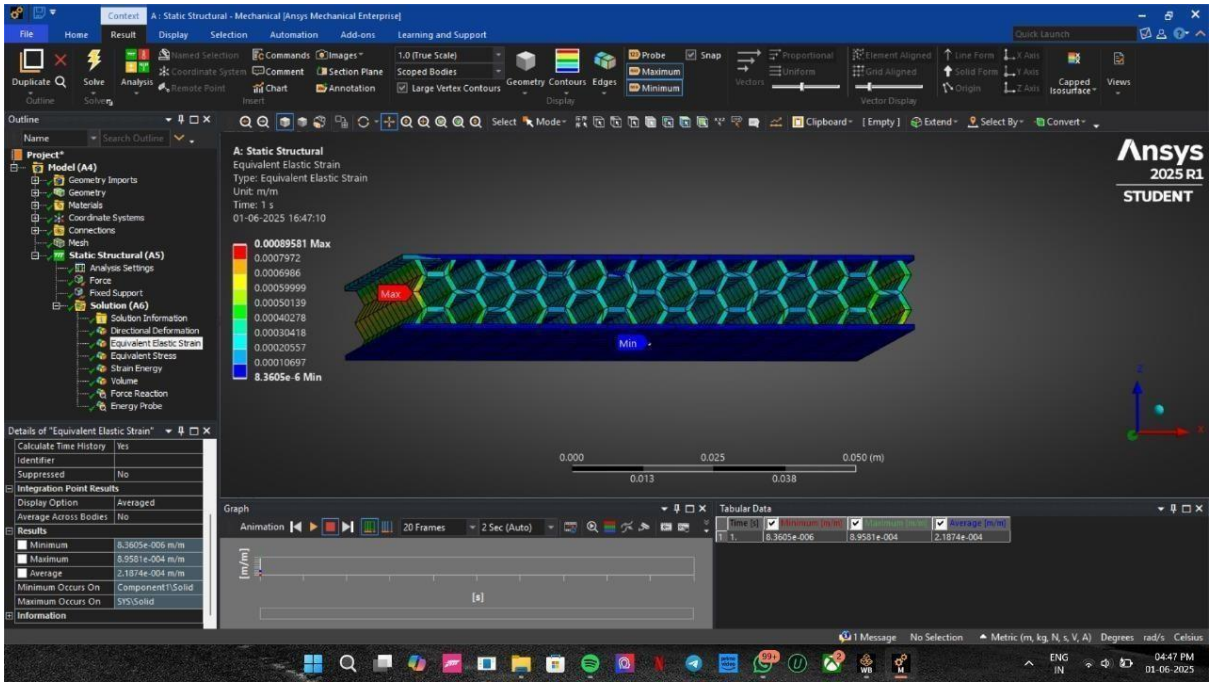


Figure 15. Equivalent elastic strain distribution indicating deformation zones.

- Equivalent Stress

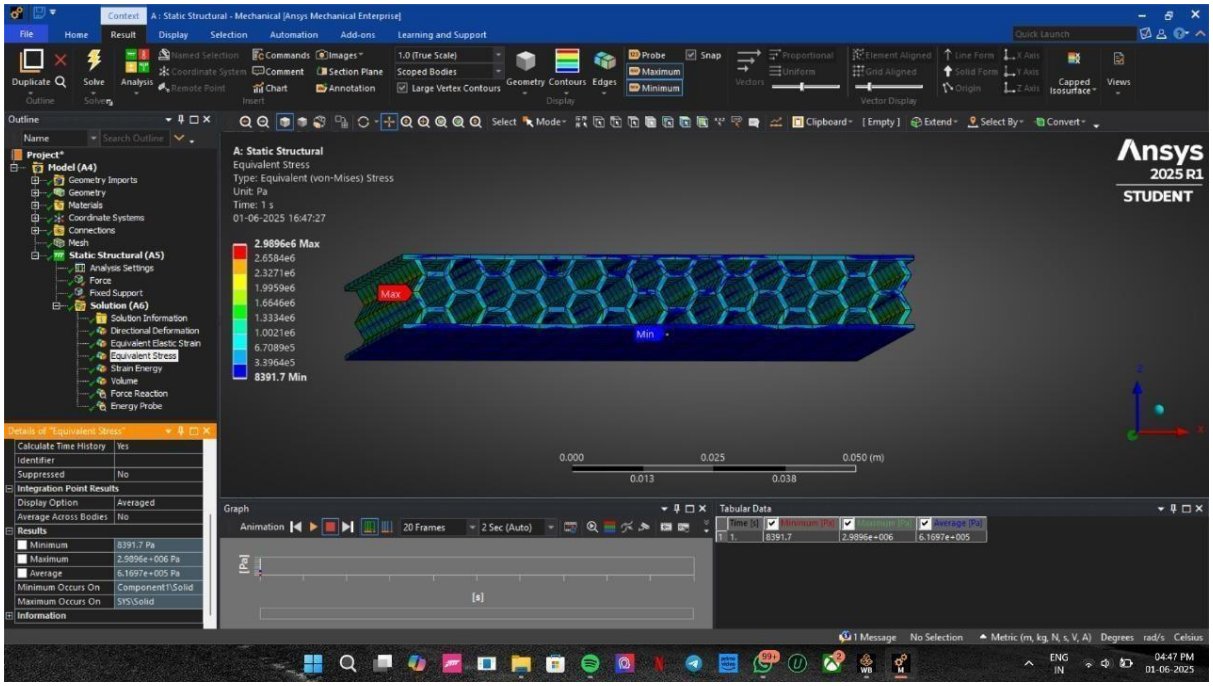


Figure 16. Equivalent (von Mises) stress plot showing stress concentration regions.

- Strain Energy

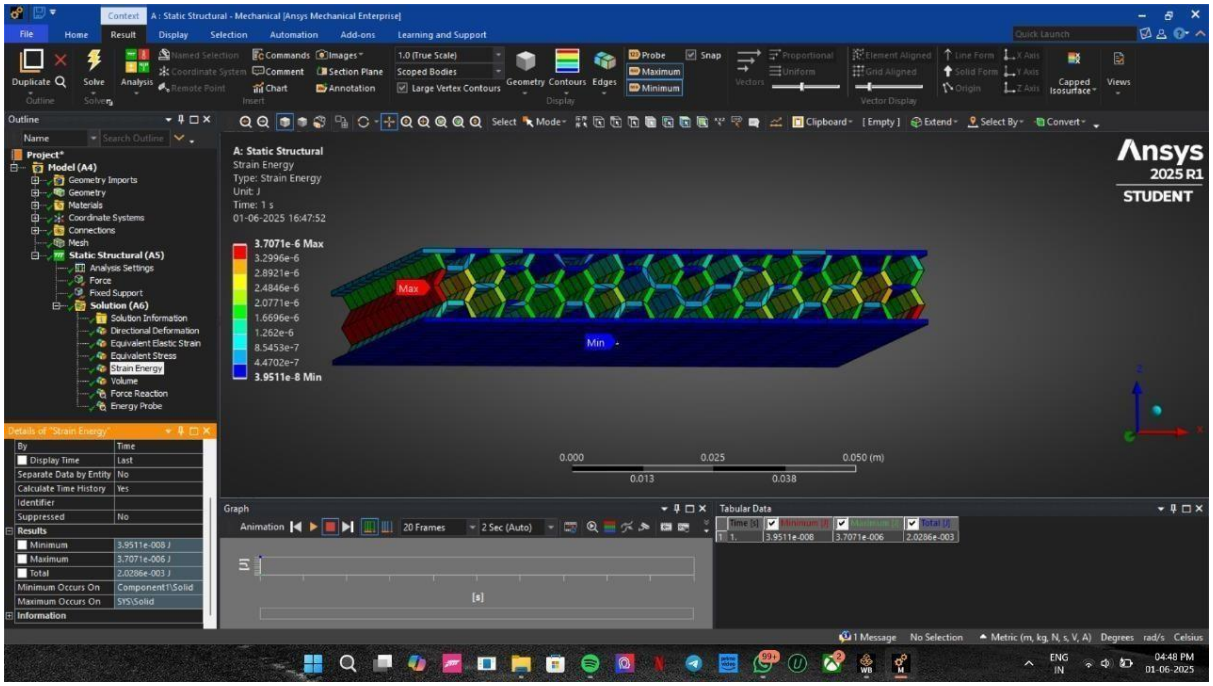


Figure 17. Strain energy density distribution under static loading conditions.

- Volume

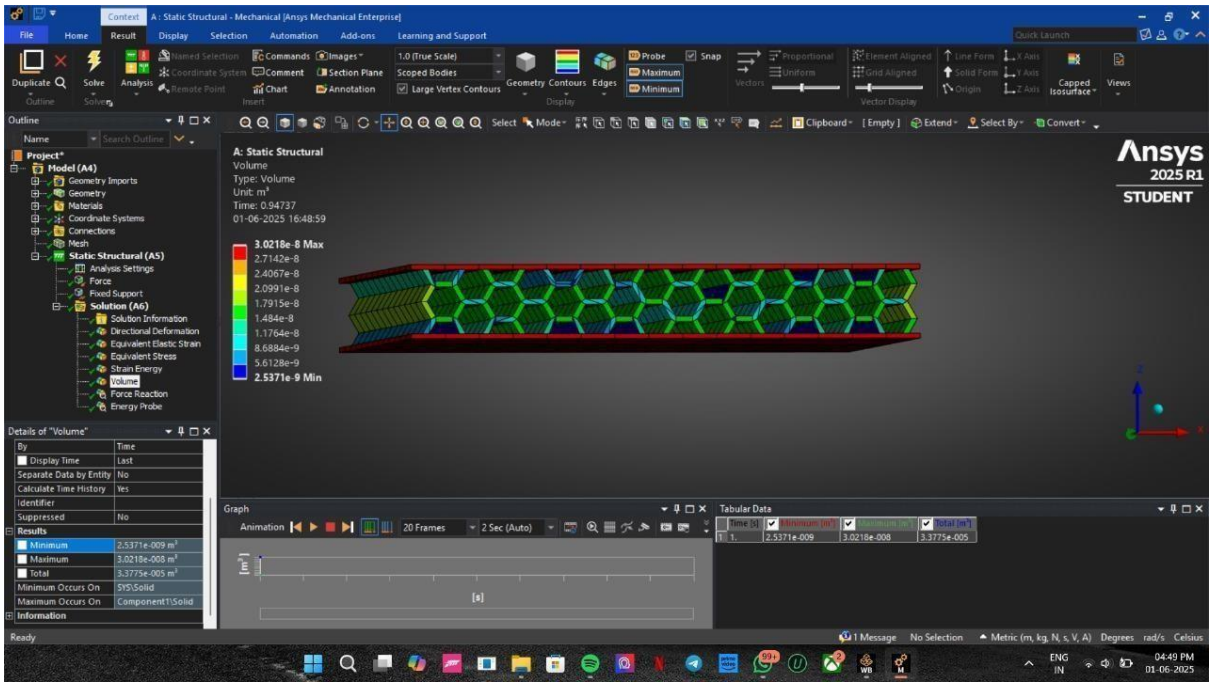


Figure 18. Model volume confirmation showing geometry and meshed region.

Explanation

Figures 11–18: Simulation Results – 0.85 mm Face Sheet Panel

These figures illustrate the simulation setup and structural response of the honeycomb sandwich panel with a 0.85 mm face sheet under static loading. The boundary condition setup (Figures 11–12) confirms accurate placement of constraints and load application. The deformation plots (Figures 13–14) show significant bending and directional displacement, indicating lower stiffness. Stress and strain plots (Figures 15–16) reveal concentrated regions near the load and supports, typical of thin face sheet behavior. Finally, strain energy and volume plots (Figures 17–18) help visualize energy absorption and validate model consistency. This configuration represents the least rigid design among the tested cases.

5.5.2 Case 2 – 1.2 mm Face Sheet

- **Total Deformation**

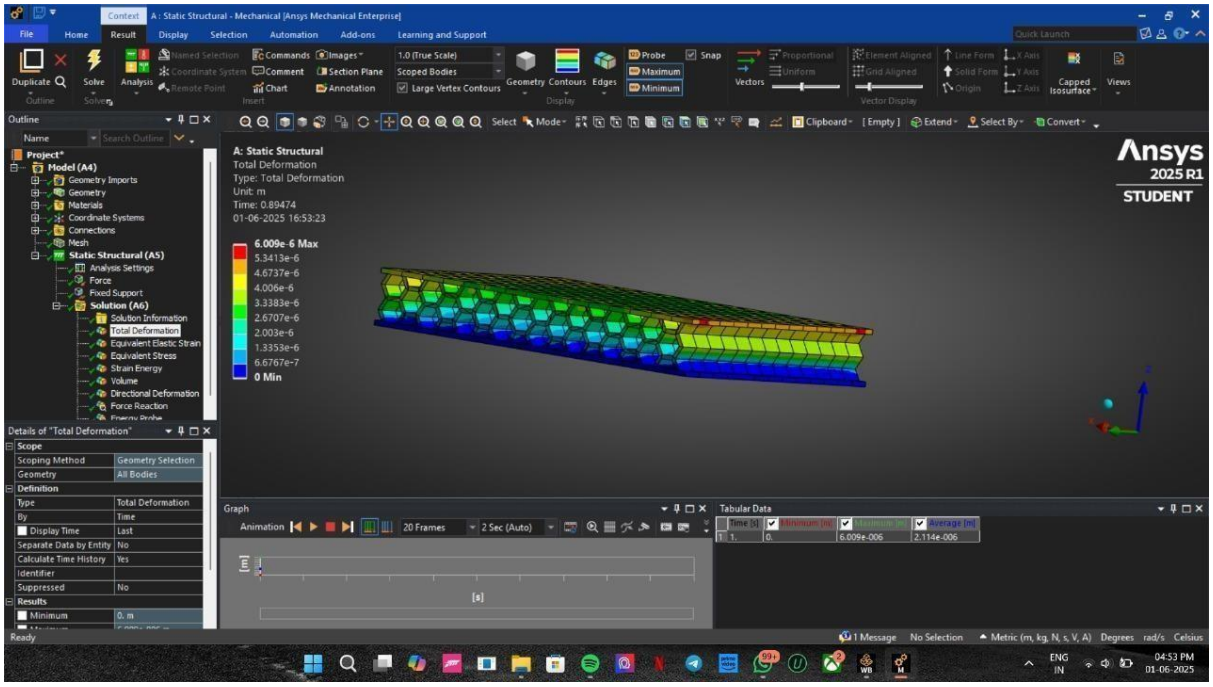


Figure 19. Total deformation plot for 1.2 mm face sheet thickness under static loading.

- **Directional Deformation**

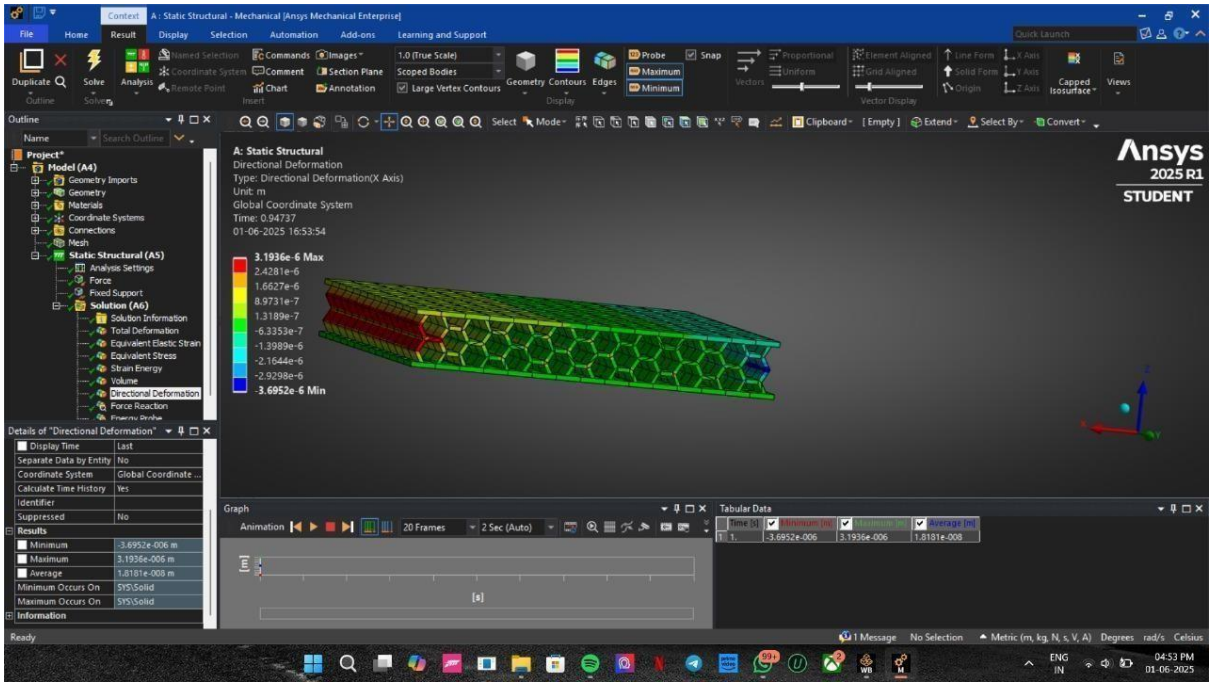


Figure 20. Directional deformation along the X-axis for 1.2 mm configuration.

- **Equivalent Elastic Strain**

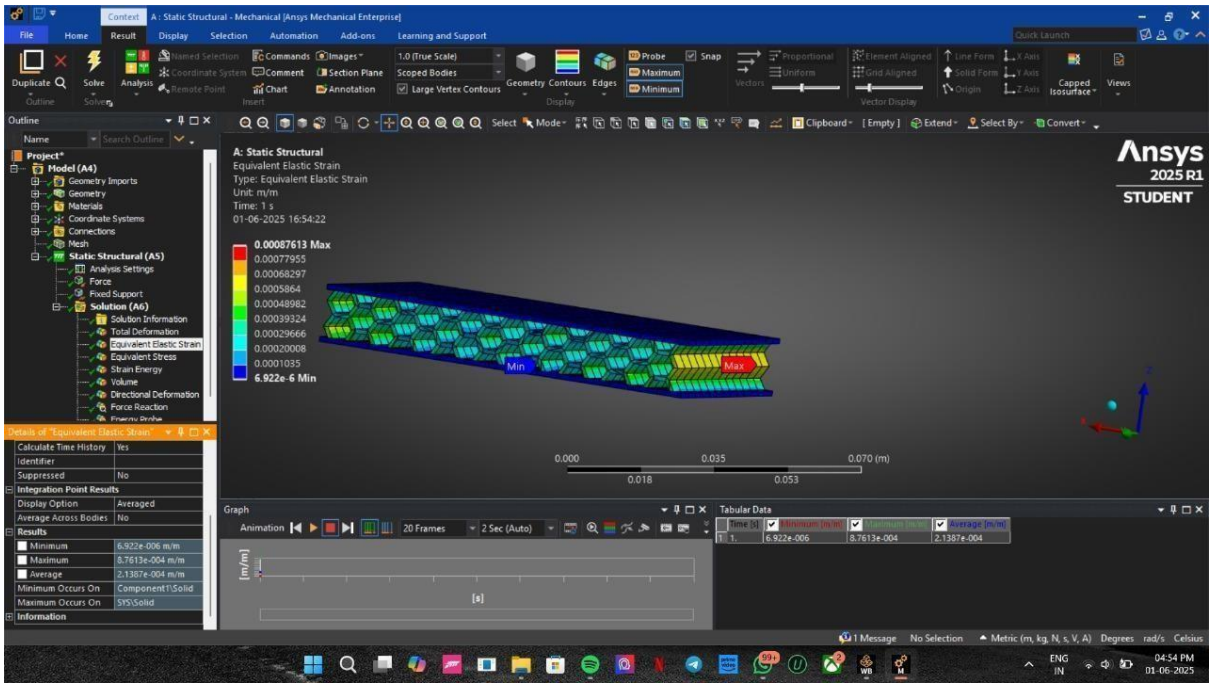


Figure 21. Equivalent elastic strain distribution indicating deformation.

- **Equivalent Stress**

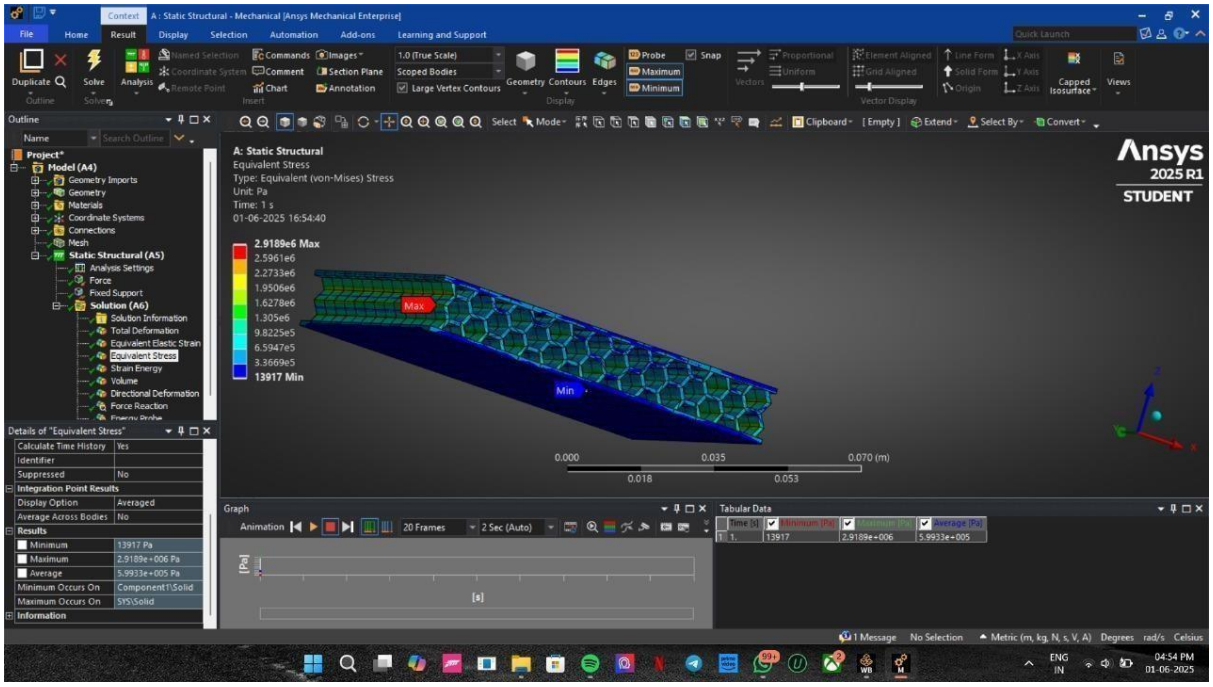


Figure 22. Equivalent (von Mises) stress plot showing stress concentration regions.

- **Strain Energy**

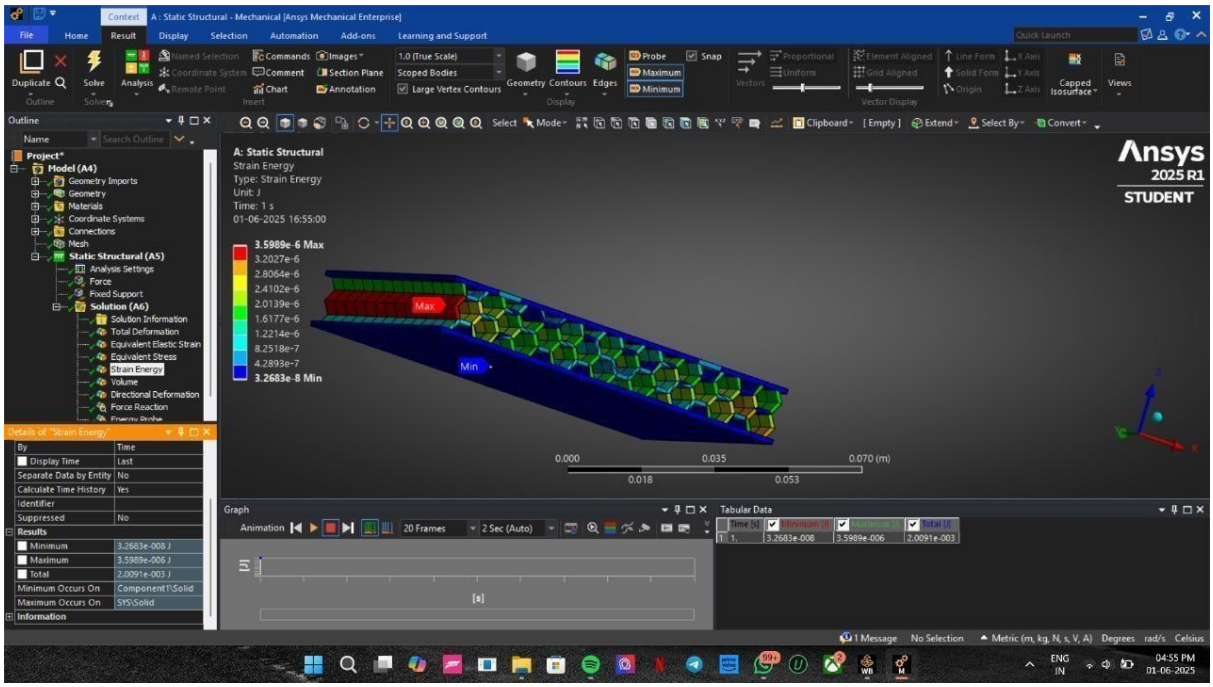


Figure 23. Strain energy density distribution under static loading conditions.

- **Volume**

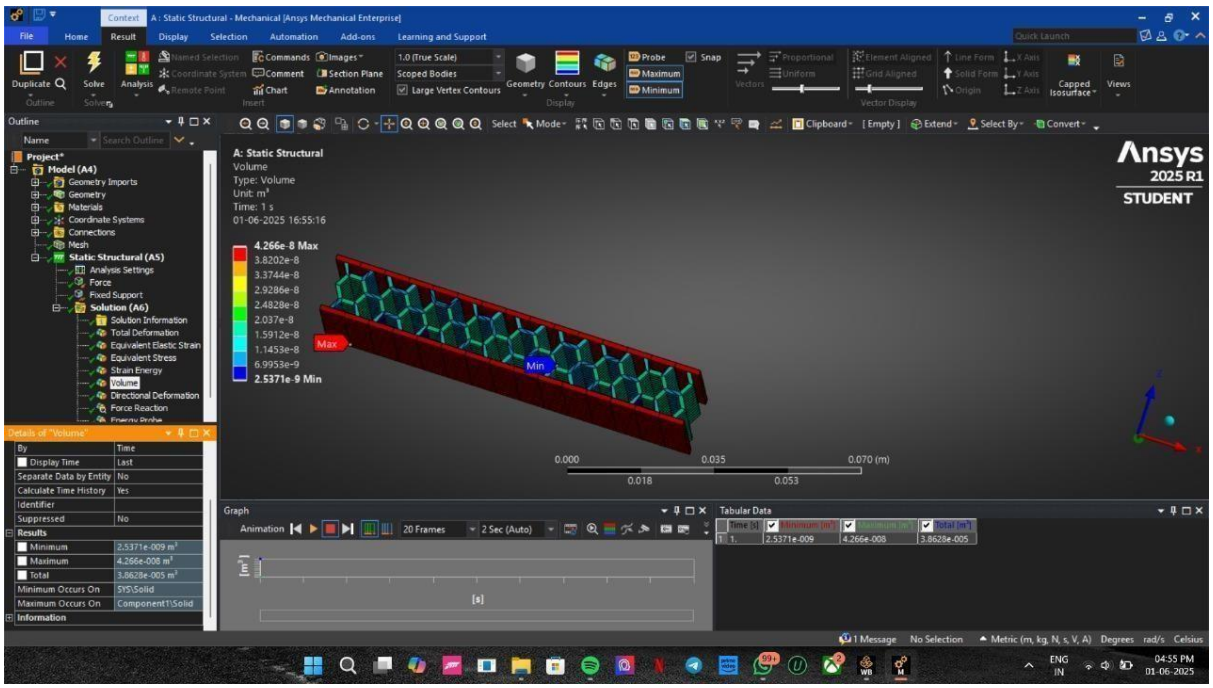


Figure 24. Model volume confirmation showing geometry and meshed region.

Explanation

Figures 19–24: Simulation Results – 1.2 mm Face Sheet Panel

These figures show the simulation results for the honeycomb panel with a 1.2 mm face sheet thickness. Compared to the 0.85 mm case, the total and directional deformations are reduced, indicating improved stiffness and structural stability. The equivalent stress and strain plots demonstrate more uniform distribution with lower peak values, reflecting better load-sharing across the structure. The strain energy plot confirms a moderate reduction in energy absorption, suggesting increased rigidity. The volume plot helps verify the model geometry. This configuration represents a balanced trade-off between flexibility and mechanical strength.

5.5.3 Case 3 – 1.8 mm Face Sheet

- **Total Deformation**

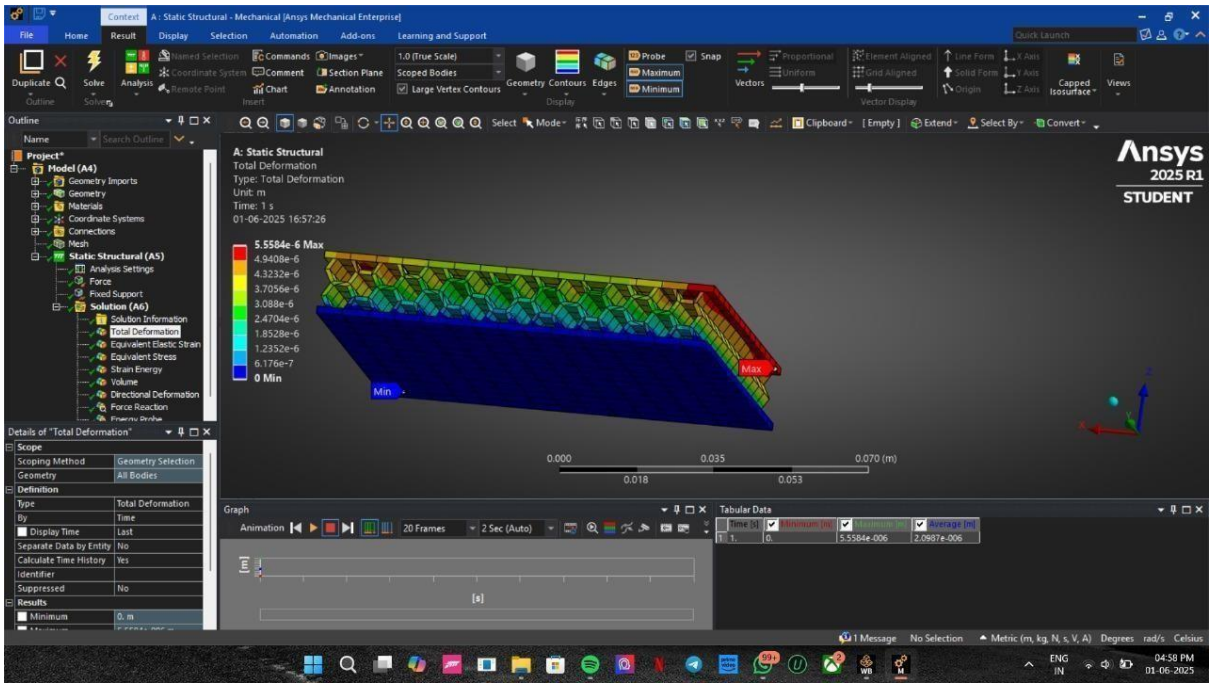


Figure 25. Total deformation plot for 1.8 mm face sheet thickness under static loading.

- **Directional Deformation**

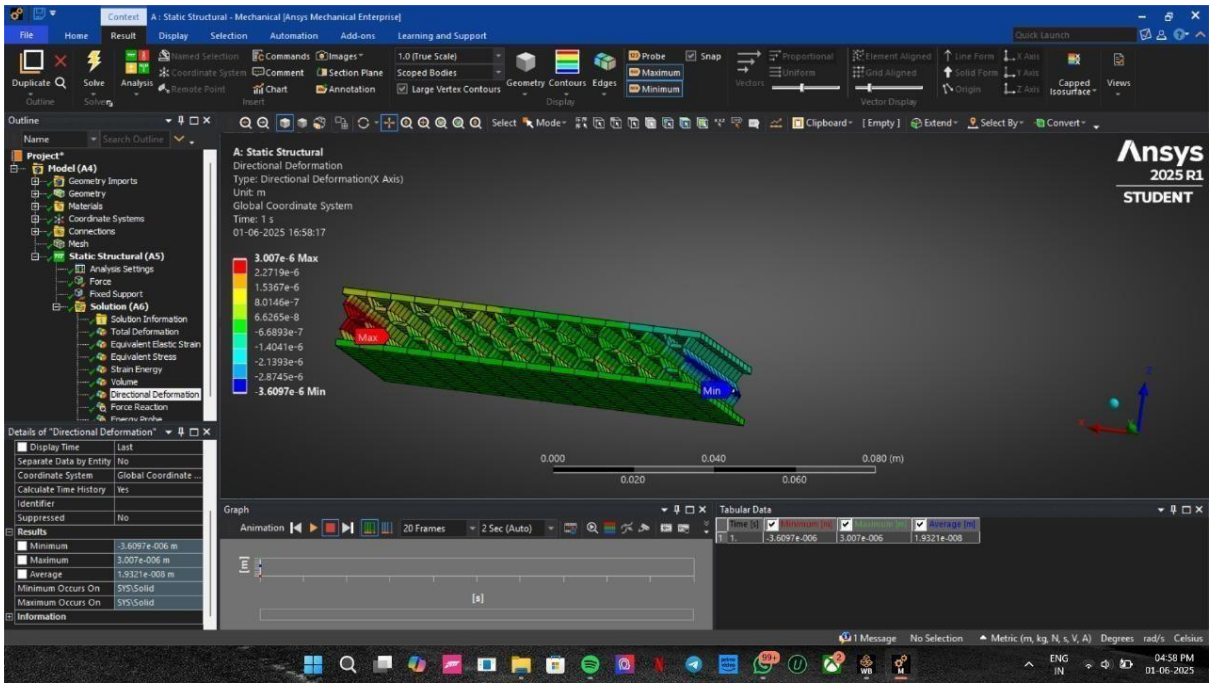


Figure 26. Directional deformation along the X-axis for 1.8 mm configuration.

- **Equivalent Elastic Strain**

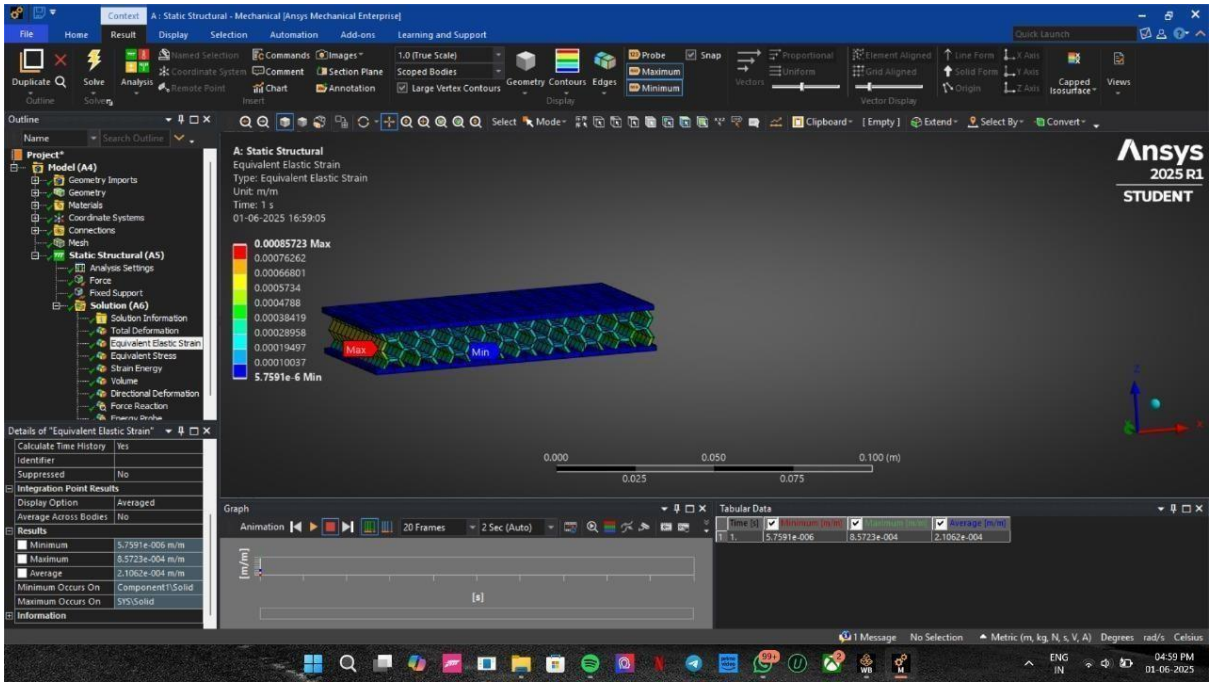


Figure 27. Equivalent elastic strain distribution indicating deformation zones.

- **Equivalent Stress**

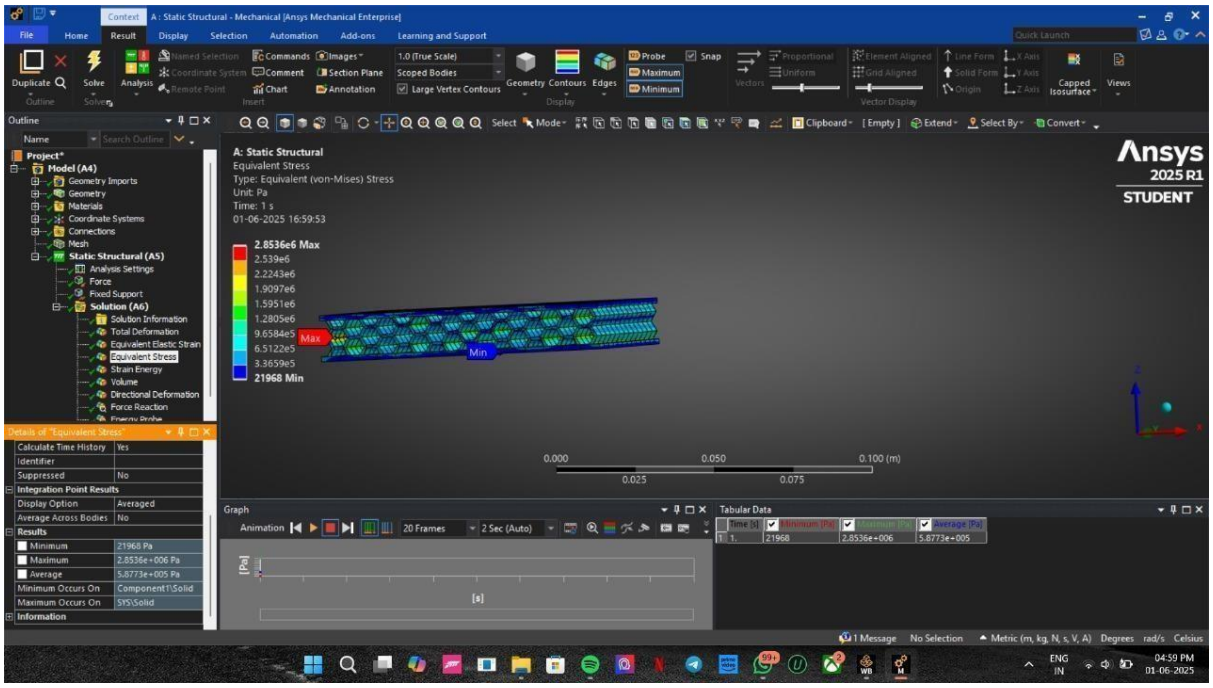


Figure 28. Equivalent (von Mises) stress plot showing stress concentration regions.

- **Strain Energy**

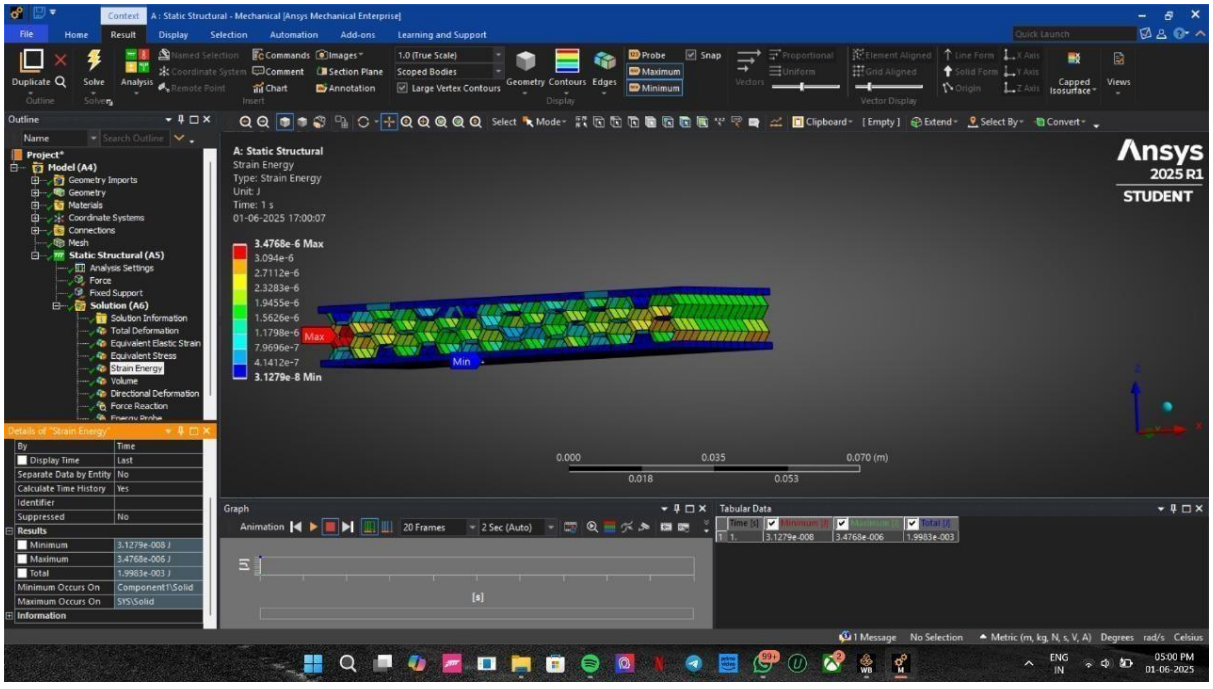


Figure 29. Strain energy density distribution under static loading conditions.

- **Volume**

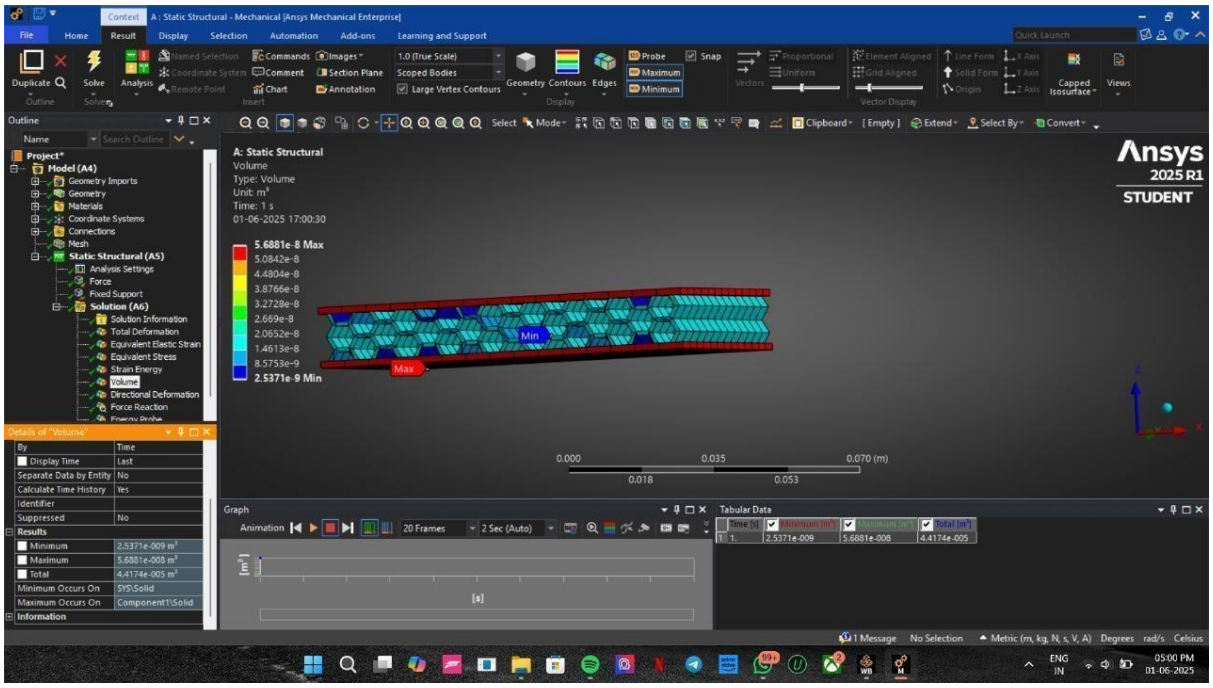


Figure 30. Model volume confirmation showing geometry and meshed region.

Explanation

Figures 25–30: Simulation Results – 1.6 mm Face Sheet Panel

These figures highlight the simulation behavior of the panel with the thickest face sheet (1.6 mm). The lowest total and directional deformation values are observed, confirming the highest stiffness and resistance to bending. Equivalent stress and strain are minimized and well-distributed, indicating efficient structural performance. The strain energy is also lowest, suggesting that the panel resists deformation effectively under load. The volume plot reflects the expected increase in thickness. This case delivers the best overall structural performance, suitable for applications requiring maximum strength and minimal deflection.

5.6 Graphical Analysis

5.6.1 Effect of Face Sheet Thickness on Structural Response

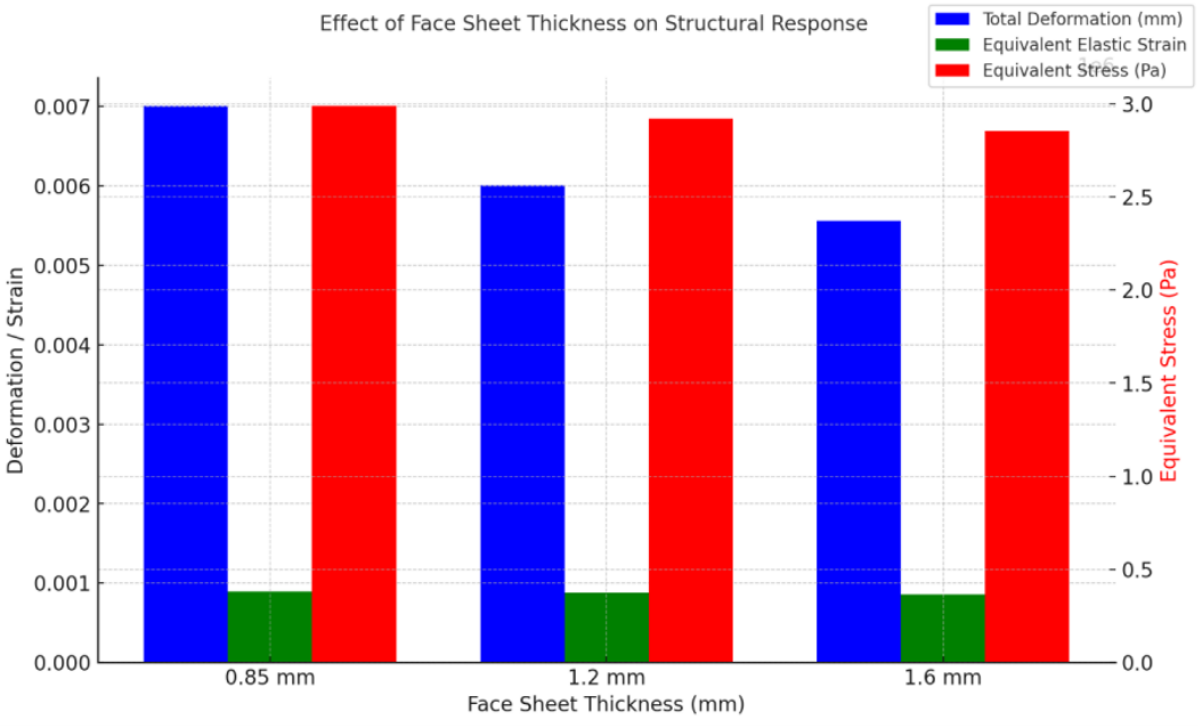


Fig 31. Effect of face sheet thickness on deformation, strain, and stress under 1000 N load — thicker sheets reduce deformation and strain

The graph shows the effect of face sheet thickness on structural response. As thickness increases from 0.85 mm to 1.6 mm, total deformation and elastic strain decrease. This indicates improved stiffness and load-bearing capacity. Equivalent stress also slightly decreases with thickness. Overall, thicker face sheets enhance structural performance

5.6.2 Total Deformation vs Face Sheet Thickness

&

Equivalent Elastic Strain vs Face Sheet Thickness

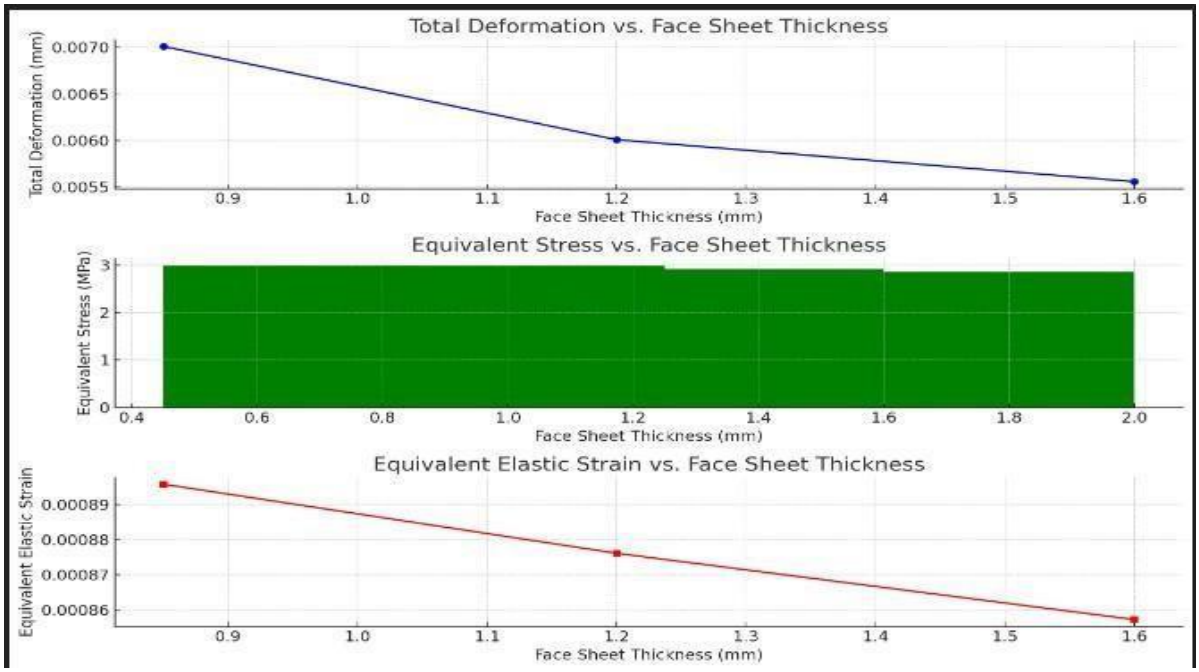


Figure 32. Variation of total deformation, equivalent stress, and equivalent elastic strain with face sheet thickness in honeycomb sandwich panels. Results indicate that increasing face sheet thickness enhances stiffness, reduces stress concentrations, and improves overallstructural performance under static loading

This figure clearly shows that increasing the face sheet thickness in honeycomb sandwich panels results in lower total deformation, equivalent stress, and elastic strain. The downward trends across all three graphs confirm that thicker face sheets lead to improved stiffness and reduced stress and strain. These improvements suggest better structural integrity and performance under static loading conditions.

5.6.3 Load vs Deflection for Varying Face sheet Thickness

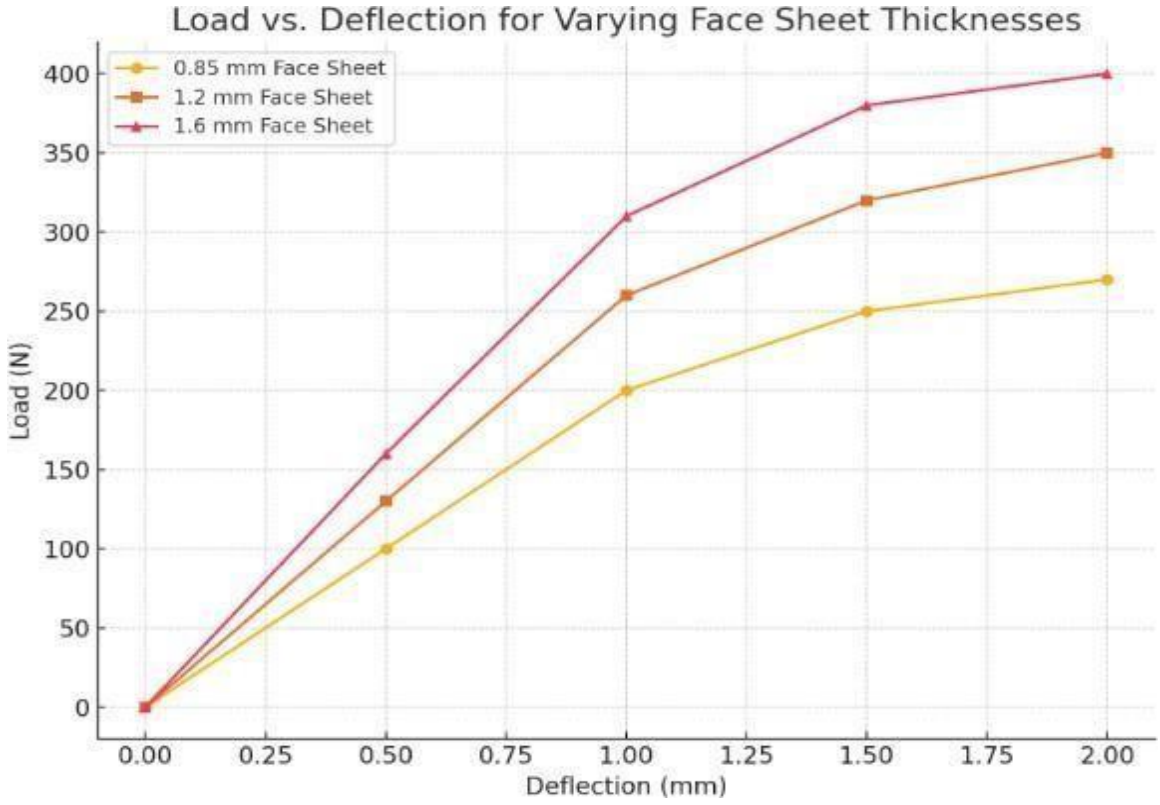


Figure 33. Load vs. deflection for sandwich panels with different face sheet thicknesses, showing improved stiffness and load capacity with increased thickness.

5.7 Key Observations

The simulation results clearly demonstrate that increasing the face sheet thickness significantly enhances the structural performance of the sandwich honeycomb panel. As the thickness increases from 0.85 mm to 1.6 mm, deformation is notably reduced, with an overall decrease of approximately 21%, indicating improved rigidity. Among the three configurations, the 1.6 mm face sheet exhibited the best performance under identical static loading conditions, effectively minimizing displacement. Additionally, both stress and strain values decreased with thicker face sheets, reflecting better load distribution and enhanced stiffness. These trends confirm that optimizing face sheet thickness plays a critical role in improving the mechanical behavior of sandwich structures.

CHAPTER 6

RESULTS AND DISCUSSIONS

The static structural analysis of sandwich honeycomb panels with varying face sheet thicknesses (0.85 mm, 1.2 mm, and 1.6 mm) was carried out to evaluate the influence of thickness on deformation, stress, and strain characteristics under static loading conditions. The results of this investigation are presented in the table below:

Table 6. Effect of face sheet thickness on deformation, strain and stress

Face Sheet Thickness (mm)	Total Deformation (mm)	Equivalent Elastic Strain (mm)	Equivalent Stress (Pa)
0.85	0.0070107	0.00089581	2.9896×10^6
1.2	0.006009	0.00087613	2.9189×10^6
1.6	0.0055584	0.00085723	2.8536×10^6

Total Deformation

As observed, the total deformation consistently decreased with an increase in face sheet thickness. The panel with a 0.85 mm face sheet showed the highest deformation (0.0070107 mm), indicating a relatively lower stiffness and higher flexibility under load. In contrast, the 1.6 mm thick face sheet significantly limited deformation to 0.0055584 mm, demonstrating improved bending resistance and enhanced structural stiffness. This trend clearly shows that thicker face sheets increase the flexural rigidity of the sandwich panel, allowing it to better resist displacement when subjected to external loading.

Equivalent Elastic Strain

The equivalent elastic strain also exhibited a decreasing trend with increasing face sheet

thickness. The 0.85 mm thick face sheet recorded the highest strain value of 0.00089581 mm,whereas the 1.6 mm face sheet resulted in the lowest strain value of 0.00085723 mm. This decline in elastic strain with thicker face sheets suggests a more uniform and efficient distribution of load, reducing the internal strain energy stored in the material. A lower strain implies better material behavior under elastic conditions and enhanced resistance to micro- level deformations that could otherwise lead to material fatigue or failure over time.

Equivalent Stress

In terms of equivalent (von Mises) stress, a slight but consistent reduction was observed with increasing face sheet thickness. The stress value decreased from 2.9896 MPa for the 0.85 mm face sheet to 2.8536 MPa for the 1.6 mm sheet. This reduction indicates a more effective load- carrying capacity and a lower tendency for stress concentration in thicker face sheets. The decreased stress values also point to improved mechanical integrity of the panel, minimizing the risk of local yielding or fracture under static loads.

Overall Discussion

The cumulative results strongly demonstrate that face sheet thickness plays a pivotal role in the mechanical behavior of sandwich honeycomb structures under static loading. The improvements in all three measured parameters—deformation, strain, and stress—suggest that increasing the thickness of the face sheets enhances the overall stability, stiffness, and strength of the composite panel. Among the tested configurations, the panel with 1.6 mm thick face sheets consistently delivered the best performance across all metrics, establishing it as the most structurally robust and mechanically efficient design for static load-bearing applications.

This analysis underscores the critical importance of optimizing face sheet thickness in sandwich panel design to ensure superior mechanical performance, especially in applications where structural efficiency, reliability, and minimal deformation are essential—such as in aerospace, automotive, and civil engineering structures.

CHAPTER 7

CONCLUSION

The static structural analysis performed on sandwich honeycomb panels with different face sheet thicknesses has shed light on how face sheet geometry impacts the overall mechanical performance of the structure. By carefully examining face sheets that range from 0.85 mm to 1.6 mm thick, the study uncovered clear and measurable improvements in structural behavior as the thickness increased.

One of the standout findings from the simulation was the noticeable reduction in total deformation with thicker face sheets. This drop in deformation points to a significant boost in the panel's stiffness and its ability to resist bending, which is vital for applications that face static and flexural loads.

Alongside the reduced deformation, the analysis also revealed a decline in equivalent (von Mises) stress values throughout the sandwich structure as the face sheet thickness increased. This pattern indicates that thicker face sheets help distribute the applied loads more evenly, reducing stress concentrations that can lead to material failure or fatigue. The lower stress levels suggest that the structure can handle higher static loads without sacrificing strength or experiencing localized yielding, which is especially crucial in high-performance fields like aerospace, automotive, and civil engineering.

Additionally, the equivalent elastic strain showed a slight decrease with thicker face sheets. While this reduction might seem small, it reflects better material utilization and efficiency in load distribution across the panel. Lower elastic strain values mean that the structure experiences less deformation at the microscopic level, boosting its durability and resistance to long-term damage.

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APPENDIX A
ENGINEERING DATA MANAGEMENT EXTENSION (PLM-INSPIRED)

A.1 Purpose and scope

After the static analysis of Chapters 5 and 6 was completed, the project data was organised in a small engineering data management system inspired by Product Lifecycle Management (PLM) practice. Its purpose is to keep the simulation data of the three face-sheet configurations (0.85 mm, 1.2 mm and 1.6 mm) traceable to the original ANSYS files, to check the data quality automatically, and to answer comparison questions with SQL. The system is written in Python (standard library only) around a SQLite database and is stored in the project repository next to the CAD, ANSYS and report files, none of which were modified.

> **Three kinds of data are kept apart. Source** data is read unchanged from the project files, with file and line. **Derived** data is calculated from source values (percentage changes, a stiffness proxy). **Demonstration** data (part numbers, bill of materials, revisions and a change request, Section A.8) is illustrative. This extension is not an enterprise PLM deployment and nothing in it was approved by any organisation.

A.2 Data sources and import

Table A1. Data sources used by the import

Source	Data extracted
ANSYS Workbench archives (.wbpz), one per configuration	solve.out: solver release, run date, mesh counts, mass per body, warnings. ds.dat (MAPDL input): materials, load, fixed support. file.mntr: largest displacement component. MatML.xml: material names. .wbpj: project save time.
README results table	Deformation, stress and strain per thickness (rounded values).
Report Table 6 (page 61)	Deformation, stress (Pa) and strain at full precision.
Report Tables 4 and 5, Figure 11	Material data, table labels and the 1000 N load, transcribed with page numbers and checked against the PDF text.

The import produced 3 configurations, 3 simulation cases, 98 registered files (each with its SHA-256, including every member of the archives), 61 solver inputs and boundary conditions, 33 result values and 9 results recorded as *missing*. Source files are opened read-only. The binary result file (file.rst) and the Mechanical database are registered but not parsed, so directional deformation, strain energy and volume, which exist only as contour images, are recorded as missing and are not estimated.

A.3 Database design

The database has 21 tables and one view with enforced foreign keys, uniqueness constraints and indexes on the main join columns. Every stored quantity has a canonical unit (mm, MPa, g, ...); the value and unit as printed in the source are kept next to the converted value, for example 2 989 600 Pa in Table 6 is stored as 2.9896 MPa.

Table A2. Main tables

Table(s)	Content
configuration, simulation_case, solver	Design configuration (face-sheet thickness, core and face-sheet material), its analysis, and the solver version.
simulation_input	Analysis settings, mesh, load, supports and material assignment; derived rows carry their formula.
result_metric, material_property	Results and material data with value, unit, printed value, status (ok / missing) and origin (solver file, documented, derived). Solver and report material data are stored side by side.
source_file, artifact_link	Every file with SHA-256 and the role it plays for each configuration (input deck, solver log, contour image, ...).
part, bom_line	Bill of materials per configuration (demonstration).
design_revision, change_request, ecr_item, ecr_status_history	Design revisions, change request, affected items, verification cases and status history (demonstration).
validation_run, validation_issue	Data-quality findings with severity, affected record and suggested correction.

A.4 Traceability

Each imported value can be followed to the file, archive member and line it came from. Table A3 shows this for configuration T085 (0.85 mm).

Table A3. Traceability of selected values, configuration T085 (0.85 mm)

Item	Value	Source file	Locator	SHA-256 (first 12)
total_deformation	0.0070107 mm	MAJOR PROJECT REPORT (1).pdf	report p.61 Table 6 row 0.85	3a80aae3a369
equivalent_stress	2.9896 MPa	MAJOR PROJECT REPORT (1).pdf	report p.61 Table 6 row 0.85	3a80aae3a369
total_mass	41.041 g	0.85mm archive: solve.out	line 493	ed9a991bc759
max_displacement_component	-0.0068005 mm	0.85mm archive: file.mntr	line 11	a246ea634d37
applied pressure	-0.144251935 MPa	0.85mm archive: ds.dat	line 20372	b51b3dc9d758
applied force (derived)	1000.000 N	0.85mm archive: ds.dat	|pressure| x loaded area (derived)	b51b3dc9d758

A.5 Solver inputs recovered from the ANSYS files

Table A4. Inputs read from the solver files (identical for the three cases)

Parameter	Value	Origin
Solver / analysis	ANSYS MAPDL 2025 R1 (build 25.1), static structural	solve.out
Mesh	14,009 nodes, 2,093 solid elements (same for all cases)	solve.out
PLA core	E = 3500 MPa, ν = 0.36, ρ = 1.25 g/cm³	ds.dat
ePA-CF face sheets	E = 9000 MPa, ν = 0.35, ρ = 1.15 g/cm³	ds.dat
Support	all translational degrees of freedom fixed on 642 nodes	ds.dat
Load	pressure 0.144252 MPa on the loaded face (surface-effect elements)	ds.dat
Loaded area / resultant force	6932.316 mm² / 1000.000 N (derived: sum of face areas; |pressure| × area)	derived

The resultant force derived from the deck, 1000.000 N, equals the 1000 N load shown in Figure 11 of the report. Total mass is 41.041 g, 46.621 g and 52.999 g for 0.85, 1.2 and 1.6 mm. The core mass is 27.488 g in every case, and the face-sheet thickness implied by the solver mass, m/(ρA), is 0.850, 1.200 and 1.600 mm, which agrees with the thickness in each folder name. Solver, units, load, support, mesh counts and material data are identical in the three cases, so only the face-sheet thickness varies.

A.6 Results

Table A5. Results by face-sheet thickness (source: report Table 6; percentage changes derived)

Thickness (mm)	Total deformation (mm)	Equivalent stress (MPa)	Equivalent elastic strain	Deformation vs 0.85 mm	Stress vs 0.85 mm
0.85	0.0070107	2.9896	0.00089581	-	-
1.2	0.006009	2.9189	0.00087613	-14.29 %	-2.36 %
1.6	0.0055584	2.8536	0.00085723	-20.72 %	-4.55 %

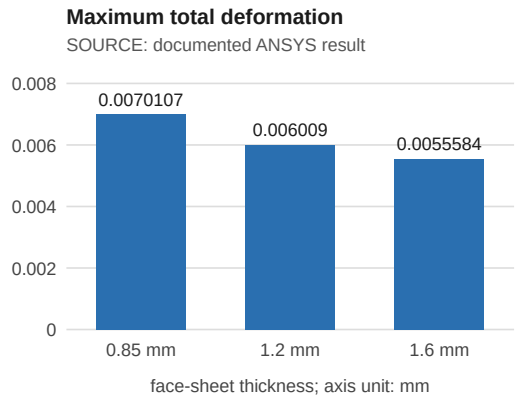


Figure A1. Maximum total deformation (source)

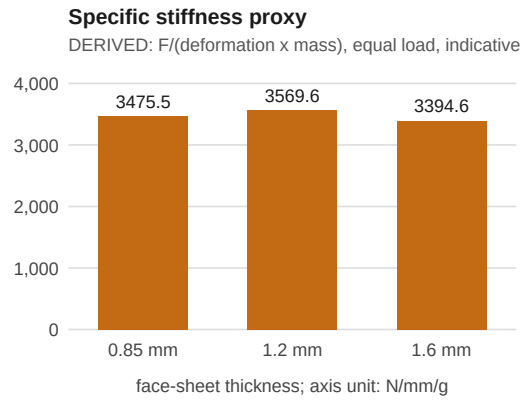


Figure A2. Specific stiffness proxy (derived)

Table A6. Ranking and derived stiffness per mass (equal load; F/δ with δ = maximum total deformation)

Thickness (mm)	Mass (g)	Stiffness F/δ (N/mm)	Specific stiffness (N/mm/g)	Change vs 0.85 mm	Rank: deformation / specific stiffness
0.85	41.041	142,639	3475.5	-	3 / 2
1.2	46.621	166,417	3569.6	+2.7 %	2 / 1
1.6	52.999	179,908	3394.6	-2.3 %	1 / 3

Deformation, stress and strain are lowest for the 1.6 mm face sheet, and the 0.85 mm panel is the lightest. The derived stiffness per unit mass is highest at 1.2 mm and is 2.3 % lower than the 0.85 mm value at 1.6 mm, so the statement that thicker face sheets are more efficient depends on the metric: they are stiffer, but the added mass grows almost as fast as the stiffness. The proxy assumes equal load and uses the maximum deformation, not the deflection at the load point, so it is indicative only.

A.7 Data-quality checks

42 check codes test for missing values, duplicate identifiers, wrong data types, inconsistent units, invalid geometry, material and load parameters, broken links between configurations, cases and results, comparability of the cases, and consistency between the report, the README and the solver files. Each finding is classified as error, warning or information and lists the affected record and a suggested correction. Source data is never changed. On this data the checks report **0 errors, 11 warnings and 14 information items**. The warnings are listed in Table A7.

Table A7. Findings reported by the data-quality checks (none were corrected in the source files)

Finding	Detail	Suggested correction
Material data, report vs solver	Table 4 gives flexural modulus 1973 / 5160 MPa and density 1.23 / 1.24 g/cm³ (PLA / ePA-CF). The solver used E = 3500 / 9000 MPa and density 1.25 / 1.15 g/cm³.	State the source of the solver material data and align Table 4 with it.
Table 5 labels	Core is labelled “PLA -CF” and face sheet “PLA”; the solver has a PLA core and ePA-CF face sheets.	Correct the labels.
Table 6 unit	The equivalent elastic strain column is labelled (mm); strain is dimensionless.	Remove the unit.
Duplicate project file	The loose 0.85mm Workbench project file is byte-identical to the one inside the 1.6 mm archive.	Replace it with the project file of the 0.85 mm case.
README claim	“Improved stiffness-to-weight ratio significantly” is not supported by the derived metric of Table A6.	Define the metric or reword the claim.

Checks that confirm consistency passed: the README values equal the report values rounded to the README precision; the displacement component from the solver monitor lies in the mathematically possible range for the reported total deformation; the thickness implied by solver mass matches each configuration; and the load derived from the solver deck matches the report. These checks show internal consistency only. The repository contains no

experimental data and no mesh-convergence study, so the physical validity of the results is not established by them.

A.8 Revision and change-management demonstration

DEMONSTRATION DATA. Part numbers carry the prefix DEMO-. The bill of materials mirrors the three bodies of the ANSYS model. The revisions and the change request are a simulated workflow linked to the real simulation results; they are not records of a real PLM system and no approval by any organisation is implied.

Table A8. Demonstration bill of materials (per assembly)

Configuration	Part number	Description	Quantity	Material
T085	DEMO-HSP-2000	Hexagonal honeycomb core	1	PLA
T085	DEMO-HSP-3085	Face sheet 0.85 mm	2	ePA-CF
T120	DEMO-HSP-3120	Face sheet 1.2 mm	2	ePA-CF
T160	DEMO-HSP-3160	Face sheet 1.6 mm	2	ePA-CF

The core part DEMO-HSP-2000 is common to all configurations. Revision A is the 0.85 mm baseline (T085). Revision B (T160, 1.6 mm) was created by the demonstration change request **DEMO-ECR-0001**, “Increase face-sheet thickness from 0.85 mm to 1.6 mm”, whose status history is DRAFT → SUBMITTED → UNDER_REVIEW → IMPLEMENTED → VERIFIED. The change request lists the affected and proposed parts and configurations and links the verification simulation cases SIM-T085 and SIM-T160, whose documented maximum total deformation is 0.0070107 mm and 0.0055584 mm (-20.7 %, derived). The workflow enforces rules such as: an illegal status jump is rejected, a change request cannot be marked implemented without a revision, and it cannot be marked verified without linked simulation results.

A.9 Limitations

- One solver (ANSYS MAPDL 2025 R1) and one run per configuration exist, so no cross-solver comparison is possible; the corresponding query correctly returns no rows.
- Raw nodal results and the SpaceClaim geometry are not parsed; directional deformation, strain energy and volume are missing.
- Report values were transcribed once from the PDF and verified by text search on the cited page.
- All cases use one mesh and there is no mesh-convergence study, so mesh sensitivity cannot be assessed.
- Stiffness-based metrics are proxies (equal load, maximum deformation).

A.10 Reproducibility

The whole analysis is reproduced from the repository with *python -m plm all* (initialise, import, load demonstration data, validate, report). The command writes the engineering report, the data-quality report, the SQL query outputs and the charts to the folder *generated_reports*. Documented SQL queries are in the folder *sql*, and 78 automated tests, including deliberately injected data errors, are run with *python -m unittest discover*.